



UNIVERSITY OF SOUTHAMPTON

2025/26

Faculty of Social Sciences
Southampton Business School

MANG6587 Capstone Project

What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

Student ID: 37595717

Presented for MSc Management and Artificial Intelligence

Supervisor: Dr Andrew Stainton
ERGO Number: 115009

September 27, 2026

This project is entirely the original work of student registration number 37595717. I declare that this submission is my own work, and that where material is obtained from published or unpublished works, this has been fully acknowledged in the references.

Word count: 16170

Abstract

Tea grading in small trading businesses is done by feel. Experienced tasters make calls that shift with the garden, the season, the buyer, and years of accumulated judgement, and no structured record of that reasoning exists anywhere.

This study asks what an artificial intelligence (AI)-enabled platform would need to do to support standardisation in that setting. The case is Shree Shyaam Tea Co., a CTC tea brokerage in Siliguri, India, with nineteen years of trading history and a single expert taster carrying the firm's operational knowledge.

Seven participants were interviewed across two stages, three from the focal company and four industry peers from separate trading businesses, selected purposively on the criterion of at least five years of tea tasting and grading experience.

Data were analysed through reflexive thematic analysis, producing six themes, and a design-validation session with the focal-company participants subsequently tested whether the proposed platform met the needs they had described.

The findings shifted the study's starting assumption. Grading instability tracks organisational boundaries rather than individual inconsistency: the same lot carries different grade conventions on each side of a transaction. Trust in AI is conditional, built through demonstrated accuracy rather than explanation, though one participant's objection is categorical in a way evidence alone cannot clear. More fundamentally, the business problem is not grading itself but scattered records, memory-based buyer tracking, and tacit expertise with no path out of one person's head.

The study contributes a proposed human-centred assessment and business support platform, grounded in both participant accounts and a case-company validation, that keeps sensory judgement with the human taster and assigns AI only to what a camera and a structured record can actually hold, what this study calls the Judgement Boundary.

Acknowledgements

This dissertation would not exist without the willingness of seven practitioners to give their time to it. Tea grading is expert work, and the people who do it are not in the habit of explaining it to outsiders. That they spoke candidly, and at length, about something they have spent years learning to do is something I do not take for granted.

My supervisor, Dr. Andrew Stainton, pushed me to be precise where I would have been vague and careful where I would have been hasty. His guidance shaped this work in ways that are visible throughout.

Shree Shyaam Tea Co. opened its doors to this research. The fact that it is my father's business made access easier; it also made the obligation to get things right feel heavier. I hope the study is worth that trust.

Contents

Abstract	i
Acknowledgements	ii
1 Introduction	1
1.1 The Business This Study Is Built Around	1
1.2 The Problem the Proprietor Cannot Solve Alone	2
1.3 What This Study Is Asking and Why	4
1.4 Why This Gap Has Not Been Closed	5
1.5 What This Study Does and Does Not Claim	6
1.6 Structure of the Dissertation	6
2 Literature Review	7
2.1 Introduction	7
2.2 Why Tea Grading Cannot Be Fully Standardised	7
2.3 What the Room Does to the Grade	8
2.4 What Computer Vision Can and Cannot Measure	9
2.5 Beyond the Camera: Sensor Fusion and Its Limits	10
2.6 Augmentation or Replacement: Where AI Should Sit	12

2.7	Why Accuracy Is Not Enough for Adoption	13
2.8	When the Expertise Lives in One Person's Head	14
2.9	What the Business Strategy Literature Offers and Where It Stops	15
2.10	What the Literature Prepared For — and What It Did Not	16
2.11	Summary	17
3	Methodology	19
3.1	Research Philosophy	19
3.2	Research Approach	19
3.3	Research Strategy	20
3.4	Research Design and Data Collection	21
3.5	Sampling Strategy and Participant Profile	23
3.6	Instrument Design and Preliminary Development	24
3.7	Data Analysis	24
3.8	Trustworthiness	25
3.9	Ethical Considerations	26
3.10	Reflexivity	27
3.11	Summary	28
4	Findings	29
4.1	Overview and Participant Profiles	29
4.2	Empirical Findings on Tea Grading, AI and Platform Requirements	30
	Theme 1: Grading Is Experiential and Context-Dependent, Not Standardised	30
	Theme 2: Trust in AI Is Conditional and Evidence-Based	31

Theme 3: AI's Realistic Value Lies in Support Functions, Not Sensory Judgement	32
Theme 4: Human Judgement and Accountability Must Remain Central	34
Theme 5: Data Control and Cost Sensitivity Shape Adoption Willingness	35
Theme 6: The Deeper Business Problems Lie Beyond Grading Itself	36
4.3 Summary	38
5 Discussion	39
5.1 Introduction	39
5.2 How the Themes Are Mapped to the Literature	39
5.3 Where the Findings Land and Where They Do Not	42
Where Grading Instability Actually Comes From	42
Conditional Trust: Explainability and the Limits of Evidence	43
Support, Not Sensory Judgement	44
Human Accountability and the Mechanisms of Oversight	45
Data Control and Cost Sensitivity Shape Adoption Willingness	45
Succession, Tacit Knowledge, and the Limits of Coordination Models	46
Where the Literature Runs Out	47
5.4 What the Themes Require the Platform to Do	48
5.5 Summary	50
6 Conclusion and Recommendations	51
6.1 What Seven Interviews Actually Found	51
6.2 Contributions of the Research	52

6.3	A Platform That Follows From the Evidence	52
6.3.1	The Physical Workflow	53
6.3.2	The Digital Platform	55
6.3.3	Rolling Out in Four Phases	56
6.4	What This Study Cannot Claim	57
6.5	What Needs to Be Built and Tested Next	58
6.6	Reflection	58
	Appendices	63
I	ERGO Ethics Documentation	64
I.1	ERGO Ethics Approval	64
I.2	Participant Information Sheet	70
I.3	Consent Form	75
I.4	Gatekeeper Letter	77
I.5	Recruitment Message	79
II	Interview Questions Asked	81
III	Company Details: Background and Preliminary Discussion	90
III.1	Company Background	90
III.2	Preliminary Discussion with the Proprietor	92
IV	Mapping of Literature to Questionnaire Design and Initial Codes	94
V	Participant profiles and key response summary	97

V.1 Participant Overview	97
V.2 Design Validation Focus Group	99
VI Full Coding Matrix	101
VI.1 Theme 1 — Grading is experiential and context-dependent, not standardised	101
VI.2 Theme 2 — Trust in AI is conditional and evidence-based	102
VI.3 Theme 3 — AI's realistic value lies in support functions, not sensory judgement	103
VI.4 Theme 4 — Human judgement and accountability must remain central .	104
VI.5 Theme 5 — Data control and cost sensitivity shape adoption willingness	105
VI.6 Theme 6 — The deeper business problems lie beyond grading itself . .	106
VII Focus Group Design Validation Evidence	107
VIII AI Declaration	110

List of Figures

1.1	Current tea assessment and trading workflow at Shree Shyaam Tea Co. Each stage relies on human judgement, manual recording, or memory. No automated step exists across the full sequence.	3
1.2	Structure of the dissertation.	6
4.1	The fixed-standard versus individual-judgement split: all three case-company participants describe grading as a fixed industry standard, while three of the four external peers rely on personal judgement and the fourth describes grade meaning as varying by garden or region.	31
4.2	Trust conditions grouped by tenure rather than company: the three participants with 15+ years' experience stack multiple conditions before trusting an AI grade, while the four under 15 years each name only one. IP4, though in the experienced group, holds a categorical objection rather than a conditional one.	32
4.3	AI's role as a spectrum rather than a hard split: five participants confine AI strictly to support functions, while P1 and E2 sit nearer the sensory edge without actually asking AI to taste or smell anything.	33
4.4	Three ways of keeping a human in control of an AI grade. "Learns from my correction" is the majority position (E1, IP1, IP2, IP3); silent override (P1, E2) and a formal flag for review (IP4 alone) are the minority positions. All seven participants keep a human in final control, but the mechanism differs.	35

4.5	Data storage preference and cost tolerance as two independent variables. IP1 is the clearest break: privacy-cautious on storage, yet the most cost-open participant, accepting a recurring subscription with no proof demanded first.	36
4.6	Tacit or hard-to-verbalise knowledge is named only by the two business owners, P1 and IP3. IP4, despite using the most vividly sensory language in the study, does not select it when asked directly.	37
5.1	Conceptual model of the proposed platform. Six features correspond to the six empirical themes. The human taster remains the decision node at every stage.	49
6.1	Single-station system with alternating 25-sample trays: operational workflow and machine architecture.	54
6.2	Simplified platform architecture for tea assessment and business operations.	56

List of Tables

2.1 Literature-Derived Design Principles and Empirical Investigation Goals .	18
3.1 Trustworthiness criteria and tactics used	26
5.1 Synthesis of empirical findings, literature connections and design implications	40
6.1 Contributions of the study	52
III.1 Company profile at a glance	90
III.2 The trading process	91
III.3 CTC grades handled	91
III.4 Themes raised in the discussion	92
III.5 Case-context codes grounded in this discussion	92
IV.1 Literature findings, key sources, initial codes and corresponding questionnaire items	94
V.1 Overview of participant roles, experience, and notable responses	97
V.2 Focus group session details	100
VI.1 Coded evidence, Theme 1	101
VI.2 Coded evidence, Theme 2	102

VI.3 Coded evidence, Theme 3	104
VI.4 Coded evidence, Theme 4	104
VI.5 Coded evidence, Theme 5	105
VI.6 Coded evidence, Theme 6	106
VII.1 Focus group responses and design validation implications	107

List of Abbreviations

AI	Artificial Intelligence
BOP	Broken Orange Pekoe
BP	Broken Pekoe
CTC	Crush, Tear, Curl
ERGO	Ethics and Research Governance Online
PD	Pekoe Dust
PF	Pekoe Fannings
SFD	Super Fine Dust
SME	Small and Medium-Sized Enterprise
TA	Thematic Analysis
TAM	Technology Acceptance Model

Chapter 1: Introduction

1.1 The Business This Study Is Built Around

Tea broking runs on fast judgement and thin margins. A broker does not grow tea or sell it under a brand name. They sit between gardens, factories and buyers, and their job is to read a tea correctly and place it before someone else does. Quality evaluation is not a step in this process; it *is* the process. A wrong grade means a wrong price, and a wrong price, often enough, means no deal at all.

Shree Shyaam Tea Co. operates out of Siliguri, West Bengal, close to India's CTC tea belt. Nineteen years of trading sit behind it. The firm takes a one percent commission from the seller on every completed transaction, which makes grading accuracy a direct commercial variable, not a quality-control aspiration. Five CTC black tea grades move through the business: BOP, BP, PF, PD and Dust/SFD. Buyers are traders, packetiers and exporters; the firm never reaches end consumers. All of it runs through one person.

That person is the proprietor, who has been tasting tea for over twenty years. He assesses dry leaf, reads liquor colour, evaluates aroma and taste, and arrives at a mark, working from memory and long experience. No written grading protocol exists. Records are held in Excel files and handwritten diaries. These work well enough day to day, but they were not built for tracing grading decisions across weeks, seasons or buyers. When a deal collapses because two parties read the same tea differently, there is nothing structured to go back to.

The gap between what the proprietor knows and what the business has formally captured is what this research is examining. A full company profile is in Appendix [III](#).

1.2 *The Problem the Proprietor Cannot Solve Alone*

Shree Shyaam Tea Co. grades the same physical lot differently depending on who tastes it, when, and under what light, and no mechanism exists to detect, record, or resolve that difference before it reaches a buyer. That is the core problem. It is not a failure of effort or skill. It is a structural feature of how grading works in this business, and, as the fieldwork shows, in the wider trade.

The owner interview identified five recurring difficulties. First, the same sample receives different scores from different tasters, with no agreed mechanism for resolving the gap. Second, lighting and weather change what the liquor looks like: a bright afternoon shifts colour perception in ways that a rainy morning does not, and the grade shifts with it. Third, buyer and trader assessments frequently diverge. The trader grades by what the tea is; the buyer grades by what they expected. Those are not always the same thing, and misalignment here is where deals break down.

Fourth, comparison across lots and across time relies entirely on memory. There is no structured record against which a new sample can be placed. Fifth, the experience that makes accurate grading possible belongs to one person. P1 said directly that the business should not depend on a single individual's accumulated judgement, but it currently does.

Current Tea Assessment and Trading Workflow (Fully Manual)

Entire process done by humans, based on experience, memory and manual work

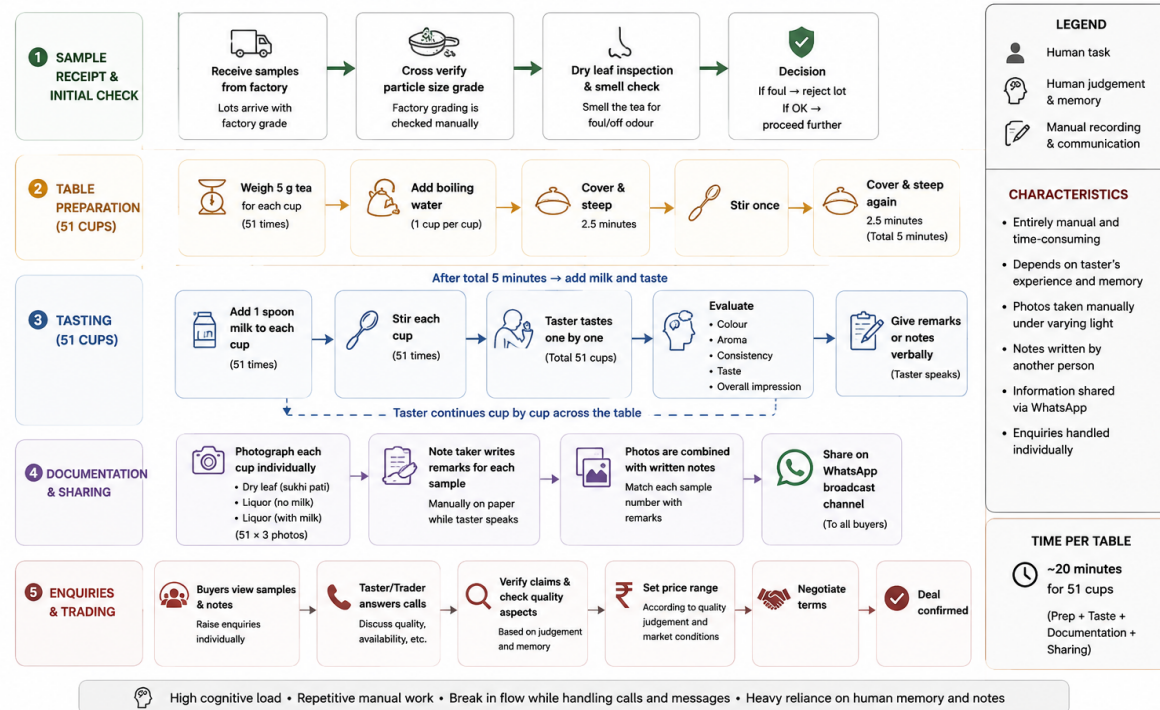


Figure 1.1: Current tea assessment and trading workflow at Shree Shyaam Tea Co. Each stage relies on human judgement, manual recording, or memory. No automated step exists across the full sequence.

Figure 1.1 maps the full workflow as it currently operates: five stages, each dependent on physical effort, taster memory, or manual documentation, with no structured mechanism connecting one stage's output to the next.

These five problems are not independent. Grading instability is partly a lighting problem, partly a calibration problem, and partly a knowledge-storage problem. Any platform that addresses only one of them will leave the others intact.

Two gaps follow directly from this: an academic gap, addressed in Section 1.4, and a practical gap facing the business itself.

1.3 What This Study Is Asking and Why

Building on the two gaps identified above, this study asks: what design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

The question grew from a practical observation. Tea quality evaluation in businesses like Shree Shyaam Tea Co. depends on trained palates, accumulated memory, and judgements that two experienced tasters can reach differently from the same sample. That is not a failure of skill. It is what sensory evaluation looks like when there is no structured mechanism to make individual reasoning visible, comparable, or transferable.

Four objectives shaped the investigation. The first was to understand current tea tasting and grading practices within a small single-expert brokerage, not how industry guidance says it should work, but how it actually does. The second was to identify sources of inconsistency in quality evaluation, whether those sit with the taster, the environment, the recording practice, or some combination. The third was to explore how participants perceive AI-assisted decision support, including where they see it as useful, where they draw limits, and what conditions they need before trusting it. The fourth was to develop design recommendations grounded in those perceptions rather than derived from technical literature alone.

Taken together, the objectives move from description through diagnosis to design. None of them assumes AI should grade tea.

The question did not arrive fully formed. An early formulation focused on tea leaf production and quality control more broadly, closer to the manufacturing literature

than the trading context that ultimately shaped the study. Engagement with the case organisation shifted that. Shree Shyaam Tea Co. is a brokerage, not a factory. Its problem is not how tea is made but how it is assessed, matched to buyers, and priced under conditions where the reasoning behind those decisions rarely gets written down. That shift moved the question away from production and towards trading businesses as its reference point.

The word standardisation entered later still, once preliminary discussion with the proprietor made clear that the real gap was not accuracy in any absolute sense but consistency, making judgements that could be compared across sessions, tasters, and time. The question as stated is the result of that sequence of corrections rather than its starting point.

1.4 Why This Gap Has Not Been Closed

Tea quality evaluation in small trading firms still runs, in most documented cases, through a single experienced taster. A fixed routine, records that live in that person's head rather than on paper, and no structured mechanism for comparison across time. The literature on AI in food quality assessment has grown considerably over the last decade, but nearly all of it addresses large-scale or laboratory settings. It assumes a technical infrastructure that firms like Shree Shyaam Tea Co. do not have ([Yang et al. 2023](#), [Wu et al. 2025](#), [Gharibzahedi et al. 2022](#)). Specifically, the literature has not examined how inter-rater, environmental, and knowledge-capture instability jointly affect grading in a single-taster trading SME, nor what such a business would want from an AI system beyond classification accuracy. That gap has not been closed.

The academic gap named in Section 1.3 is not a call for another study of classification accuracy under controlled conditions. It is a call for a grounded account of the practices and priorities of the people who would use such a system. Research on technology acceptance in small firms ([Hossain et al. 2024](#), [Ikpe 2024](#)) and on human-in-the-loop AI design ([Lazaros et al. 2026](#), [Ma et al. 2023](#)) rarely reaches this far into the value chain. When it does, the business in question is larger, better resourced, and not dependent on one person for every grading decision made.

The practical gap is more immediate. Grading decisions at Shree Shyaam Tea Co. rest on one person. If that person is unavailable, or eventually leaves, there is no structured record of how those decisions were made. This is not unusual in the sector. It is the kind of institutional fragility that a well-designed platform could, in

principle, address. Whether that is what participants actually want is what this study is built to find out.

1.5 What This Study Does and Does Not Claim

The study identifies what such a platform would need to do rather than building a working system. No prototype is constructed, no technical performance is measured, and no claim is made that the proposed design would improve grading accuracy or commercial outcomes. Testing those claims requires building something first, and that is a different project.

The empirical base is a single tea brokerage in Siliguri and seven participants across owner, employee, and industry peer roles. That sample size was realistic, not a compromise. Tea grading is expert work, and access to practitioners willing to discuss it in detail is narrow by nature. Seven participants gave genuine variation in experience and role. What seven people cannot give is breadth. The findings describe Shree Shyaam Tea Co. and, more tentatively, comparable small CTC brokerages. They do not describe the Indian tea trade as a whole.

What this study produces is a grounded account of what one real business needs from such a system. Not a proof that it works.

1.6 Structure of the Dissertation

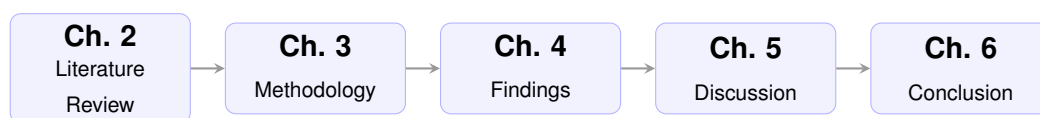


Figure 1.2: Structure of the dissertation.

Figure 1.2 summarises this flow. The dissertation moves through five substantive chapters. Chapter 2 reviews the literature across sensory evaluation, computer vision, human-in-the-loop design, technology acceptance, and knowledge management in SMEs. Chapter 3 sets out the research design: a two-stage qualitative approach combining a self-administered questionnaire with semi-structured interviews at Shree Shyaam Tea Co. Chapter 4 presents the findings as six empirical themes. Chapter 5 discusses those themes against the literature. Chapter 6 draws conclusions and offers recommendations.

Chapter 2: Literature Review

2.1 Introduction

Tea quality assessment sits at an odd crossroads. The trade still leans on trained palates and personal judgement, yet the wider food and beverage sector has moved towards automation to raise efficiency and reduce disruption to production, [Ouvinha de Oliveira et al. \(2021\)](#). Predictive, data-driven decision-making has grown alongside this shift, and within tea specifically, faster and more reliable evaluation is increasingly expected as quality and safety standards tighten ([Wu et al. 2025](#)). This chapter works through the literature that bears on this problem.

A distinction matters here, and it shapes how the rest of the chapter should be read. Standardisation and automation are not the same goal, though the literature often treats them as though they were. Automation asks a system to produce the grade. Standardisation asks a system to make the reasoning behind a human grade more consistent and visible, while leaving the decision itself with the person who already makes it. This chapter is positioned closer to the second goal, and the gap in Section [2.10](#) follows from it.

2.2 Why Tea Grading Cannot Be Fully Standardised

Tea quality assessment rests on human judgement, and human judgement is not a stable instrument. A tea taster works through a fixed sequence: dry leaf, liquor colour, aroma, taste, infused leaf, then assigns a mark, usually on a scale of one to ten ([Bhattacharyya et al. 2008](#)). The scale looks objective on paper. In practice, the same taster applying the same scale to the same tea, on a different day or under different conditions, does not always reach the same number.

[Bhattacharyya et al. \(2008\)](#) describe human panel tasting as subjective, prone to inaccuracy and non-repeatability, and dependent on factors unrelated to the tea itself:

the taster's experience, prior exposure to strong aromas, health, or general state of mind. None of these factors appear in the final grade.

[Sajeewani et al. \(2023\)](#) frame the issue differently. Professional tasters, they note, categorise tea quality in a subjective manner, weighing several parameters at once, of which flavour, aroma and colour carry the most weight. Their more useful claim is that these subjective judgements correlate with the tea's underlying chemical composition, particularly flavanol content. The taster is not inventing the quality difference. They are perceiving something genuine, then translating it into a number through a process that is not itself consistent from one sitting to the next.

[Laddi et al. \(2012\)](#) trace the inconsistency further. Human organoleptic methods, they argue, struggle to discriminate between teas close in colour, odour or particle size, and the resulting errors relate to differences between panellists, to subjective interpretation, and to sensitivity dulling after prolonged exposure to successive samples. This last point is relevant to a single-taster operation, where fatigue across a working day is a structural possibility rather than an occasional risk.

[Guo et al. \(2025\)](#) summarise the field's general position: sensory evaluation and physical inspection, used as quality control methods, tend to lack objectivity and fairness, which is part of why computer vision and chemical-analysis alternatives have attracted research attention. The point running through these papers is narrower than a criticism of any individual taster's skill: subjectivity in tea tasting is presented in this literature as a structural feature of asking a human sensory system to produce a repeatable number, not simply a flaw that training removes.

That framing already implies something about what a platform should do: if subjectivity is structural rather than incidental, a system aimed at removing it entirely through automation is aiming at the wrong target, whereas a system aimed at making the structure visible and comparable, standardisation in the sense set out in [Section 2.1](#), is aiming at a more realistic one. [Section 2.3](#) turns to a second, related source of inconsistency: the physical environment in which that judgement is made.

2.3 What the Room Does to the Grade

Subjectivity in tea evaluation does not sit only inside the taster's head. It comes from the tea itself and from the conditions the taster works in. Sensory science treats these as separate but linked sources of variance, and tea research bears that out.

Wu et al. (2025) note that tea's sensory attributes, including colour, flavour and aroma, are shaped by region, climate, cultivation and processing before a taster ever sees the sample. Laddi et al. (2014) extend this with evidence from black CTC tea, showing that theaflavin and thearubigin content, the compounds most responsible for liquor colour and taste, shift measurably across harvesting seasons, alongside atmospheric conditions, soil fertility, altitude, rainfall and manufacturing technique. A tea assessed in March is, chemically, not quite the same product as one assessed in July, even from the same garden.

If the object being judged changes in ways a taster cannot always see or name, then asking that taster to produce a stable score becomes structurally difficult, not simply a matter of personal inconsistency. The literature reviewed here points specifically to lighting and weather as sources of visual variation in tea appraisal. Inconsistent or poor lighting degrades image quality and undermines the reliability of electronic-eye and camera-based assessment tools, which depend on stable conditions to read colour and brightness accurately (Zhi et al. 2025). Environmental inconsistency, then, sits alongside inter-rater variability as a second, largely separate source of noise in tea grading. Whether and how this plays out at Shree Shyaam Tea Co. specifically is a question for the empirical chapters rather than the literature itself.

2.4 What Computer Vision Can and Cannot Measure

Computer vision has become a widely used non-destructive method for external quality assessment in food and agriculture, valued for its accuracy, speed, and the volume of data it can process once a system is built (Zhi et al. 2025). Machine vision follows a broadly similar pipeline across many agricultural applications, tea included: an image is acquired under controlled lighting, features are extracted, and a model classifies the result (Yang et al. 2023). What varies between studies is which features actually carry the grading signal, and the tea literature offers findings that do not fully agree.

Rahman et al. (2021) built a machine vision rig for four CTC grades and extracted physical parameters: area, perimeter, circularity, feret diameter, aspect ratio and solidity, from thresholded images taken inside a light-controlled box. Area, perimeter and feret diameter separated the grades with strong statistical significance ($p < 0.01$). Aspect ratio and solidity showed no significant difference between grades at all. Gill et al. (2013) took a different approach on a different set of black tea grades, extracting statistical texture descriptors from grey-level co-occurrence matrices, then repeating

the exercise on wavelet-decomposed sub-bands. A multilayer perceptron reached 80% accuracy on the plain statistical features and 82.33% once wavelet features were added.

Both studies share a similar limitation: grade pairs close in granule size and shape were harder to separate reliably using either texture or geometry. This raises a relevant question for Shree Shyaam Tea Co., where PF and PD grades occupy relatively close positions on the particle-size scale. Whether this creates a comparable classification challenge in practice is a matter the empirical data can speak to, not something the literature alone can establish.

The wider tea literature has moved towards deep learning, and [Yang et al. \(2023\)](#) survey the shift: shape-histogram classifiers reaching over 95% accuracy on black tea grading, attention-based networks reaching close to 100% on some price grades. The same review is more cautious about how narrow the underlying datasets tend to be. Models are usually trained on one variety, one region, one season, often one fixed lighting rig, with no shared benchmark across the field. A model that performs well on a bench under steady artificial light is not necessarily the model that would perform equally well under variable natural daylight in a working trading environment.

[Zhi et al. \(2025\)](#) note that computer vision, however precise, reads only what is on the surface, colour, shape, texture, and cannot capture the chemistry that also determines a tea's value: the polyphenols, amino acids and caffeine that shape strength and character in the cup. A camera can grade the leaf. It cannot taste the liquor.

Taken together, this section's literature demonstrates that computer vision can measure specific observable characteristics of tea reliably under controlled conditions, but struggles precisely where grading distinctions are finest and where conditions are least controlled, the situation closest to Shree Shyaam Tea Co.'s working environment. The implication is not that computer vision has no role, but that its role is narrower than full automated grading: it is better suited to standardising the measurable, visible component of an assessment than to replacing the assessment itself.

2.5 Beyond the Camera: Sensor Fusion and Its Limits

One response to the limitations of visual assessment has been the development of sensor-based systems designed to capture characteristics associated with aroma, taste and appearance more directly. [Gharibzahedi et al. \(2022\)](#) provide an overview

of this field, covering three device families: the electronic nose, which profiles volatile aroma compounds; the electronic tongue, which performs a comparable function for taste; and the electronic eye, which measures colour and appearance through image analysis. No single sensor captures the full picture, the authors note, since aroma, taste and colour interact in ways one device cannot register alone.

[Ren et al. \(2023\)](#) demonstrate what combining sensors looks like in practice. Working with seven grades of Congou Dianhong black tea, they paired a portable near-infrared spectrometer with an electronic tongue and reached a classification accuracy of 93.56%, an improvement on either sensor used alone. [Peng et al. \(2025\)](#) take a more taster-centred approach: rather than replacing the taster, they trained a model to correlate spectroscopic readings of nine chemical constituents with grades assigned by professional tasters, reaching 95% agreement with the human benchmark. The system was explicitly framed as reducing the taster's workload rather than removing the taster from the process.

These systems perform well under controlled conditions. Each requires specialised sensing equipment, spectrometers, electrode arrays, sensor chambers, that a small single-taster CTC brokerage is unlikely to have access to or be able to justify acquiring in the near term. And none of the three studies removes the human taster from the process: each treats sensor output as something to train against or fuse with professional judgement, not as a replacement for it.

The pattern across this section and the previous one is consistent and worth stating directly. Neither computer vision nor sensor fusion, across every study reviewed here, produces a system designed to operate without a human taster in the loop, and neither is designed for deployment in a resource-constrained single-expert setting.

The implication for platform design is twofold: full technical automation of tea grading is not what the reviewed literature is building towards, even where it could technically attempt it, and any platform for a business like Shree Shyaam Tea Co. will need to work with a much lighter technical footprint than the equipment described in this section assumes. That reinforces, rather than undermines, the standardisation framing set out in Section 2.1: the relevant question shifts from what AI can measure to how AI and taster should work together, which needs a different body of literature entirely.

2.6 *Augmentation or Replacement: Where AI Should Sit*

Automation and augmentation describe two different things AI can do to a task, and [Schröder et al. \(2022\)](#) draw the line clearly: automation means the machine takes over what a person used to do, augmentation means person and machine do the task together. This is the same split raised in Section 2.1 under different names, standardisation versus automation, just borrowed here from the decision-support literature. For tea grading the distinction is not academic. It decides whether a system replaces the taster's judgement or backs it up. Automating a task with a clear right answer is one problem. Automating a task where the outcome is a call shaped by memory, experience, and context is a different problem entirely, and Section 2.2 already placed tea grading in that second category.

[Ma et al. \(2023\)](#) give a good reason to be careful with the automation route. Their study of AI-assisted decision making found that people who receive AI recommendations passively rarely think them through. They accept or reject, and that is it, no real reasoning in between. Apply that to a taster leaning on an algorithm's grade, and a real question opens up: is the tool actually supporting the decision, or is it just swapping one form of pattern matching, a trained palate, for another, a model's parameters, while looking safer because a human still clicks "confirm"? A standardisation-minded design tries to avoid exactly this. Built to surface comparisons and evidence rather than spit out a grade, it gives the taster something to think with instead of something to defer to.

[Lazaros et al. \(2026\)](#), reviewing human-in-the-loop systems, argue current AI still cannot do what tea grading demands: read context, weigh relational factors, be accountable for a call that affects a real transaction. [Retzlaff et al. \(2024\)](#) add that incorporating even imperfect human input outperforms ignoring it: agents that disregard human advice, however suboptimal, tend to learn worse policies than agents that use it.

There is a related question about how experts work with a highly autonomous system when the task is creative rather than computational. [Tchemeube et al. \(2025\)](#) studied a co-creative AI music tool inside a professional composing workflow. Expert users rated it usable and mostly accepted it, but kept asking for finer control rather than one blunt parameter governing everything. Composers said they felt less like authors when they could not see how the system generated material, and expressiveness, the lowest-rated dimension, tracked with that lack of visibility.

Music and tea grading are not the same domain, so this should not be stretched too far, but it hints at a pattern worth testing in tea: experts may put up with an AI system longer if they can see roughly what it is doing and step in when they disagree.

Put together, this literature points to one conclusion. AI in judgement-heavy work does best as reference and comparison, not verdict, because automation strips away the visibility and interrogability that human-in-the-loop research treats as necessary. The risk Ma et al. describe, passive acceptance wearing down the judgement a system is meant to support, becomes the design constraint for the next section: trust depends on actually understanding what a tool is doing, which is where explainability comes in.

2.7 Why Accuracy Is Not Enough for Adoption

Adoption of an AI tool rarely comes down to accuracy alone. It comes down to whether the person using it is willing to trust what it tells them, and this is where TAM becomes useful, not as a stand-alone model but as one of three lenses working together: human-in-the-loop asks who decides, explainability asks how AI should support that decision, and TAM asks why the decision maker would adopt the system in the first place.

[Sohn & Kwon \(2020\)](#) describe TAM as growing out of the Theory of Reasoned Action, built to explain user acceptance of information systems, and note it remains the most widely used model for this purpose. The model itself dates to [Davis \(1989\)](#), who proposed perceived usefulness and perceived ease of use as the two constructs governing whether a user adopts a new system, a framework later work has extended rather than replaced. [Gefen et al. \(2003\)](#) extend the model by adding consumer trust as an independent predictor of intended use alongside perceived usefulness and ease of use, finding trust nearly as influential as the standard TAM antecedents, though perceived usefulness remains the stronger of the two.

That distinction matters here. In tea grading, there is already an established source of trust: the taster's own palate. Any AI system entering that space competes with a working baseline rather than filling an empty one.

The trust question gets harder once the system in question is a black box. [Gamoura \(2023\)](#) makes the point plainly: AI's inductive, opaque nature, combined with implicit execution, breeds reluctance and trust problems on its own. [Abayomi & Noordeen \(2025\)](#) arrive at something similar from a different angle, observing that organisations

frequently deploy AI without the explainability frameworks stakeholders would need to actually question a decision, leaving them stuck with outputs they cannot interrogate.

The picture is more conditional than pessimistic. [Corvelo Benz & Gomez Rodriguez \(2023\)](#) complicate this further, showing that well-calibrated confidence values are not in themselves enough for a decision maker to work out how much to trust a prediction; monotone trust only becomes discoverable when those values are aligned with the decision maker's confidence in their own judgement. [Chang & Bau \(2025\)](#) tie the threads together by defining explainable AI as a framework that lets users understand how a process reached its output, which in turn raises the reliability people are willing to assign to the result.

Two things follow. First, ease of use and comprehensibility are not the same thing, and a system can be simple to operate while still being hard to trust. That is a distinct failure mode, and any platform built for Shree Shyaam Tea Co. would need to design against it directly. Second, and more central to the standardisation argument this chapter is building towards, explainability is what keeps a standardisation-oriented system from quietly becoming a replacement grader. An opaque number, however accurate, behaves identically whether it was meant as a decision aid or a substitute for the taster, because the taster has no way to tell the two apart. Visible reasoning is what keeps that difference real rather than nominal.

2.8 When the Expertise Lives in One Person's Head

Tea grading expertise built over decades currently lives in one person's head. It has never been written down in any structured way. This is the problem the dissertation keeps returning to, and it matters more than it might first appear. A platform built purely to standardise grading addresses something real, but narrow: consistency in the moment. A platform that also captures the reasoning behind years of individual grading decisions addresses something else entirely, institutional memory in a business whose core asset is one irreplaceable person. That reframes the whole project. It stops being AI-enabled grading and becomes something closer to AI-enabled knowledge preservation, with standardisation as one output rather than the point of the exercise.

[Wu et al. \(2025\)](#) identify a related problem at industry scale. Even where deep learning models for tea quality already exist, the sector lacks enough skilled, well-trained professionals to formalise or transfer that expertise through technology in

the first place. This is not a small gap. [Ikpe \(2024\)](#) helps explain why a small trading firm in particular would struggle here, even with the motivation to try. Small businesses in developing-economy contexts tend to face limited resources, limited technical expertise, and worries about job displacement when adopting AI, and these constraints stack on top of an already thin pool of AI professionals available to support them.

There is also an organisational angle worth noting. [Baumann & Wu \(2023\)](#), writing in a special issue on flatter, less hierarchical organisational design, suggest AI-enabled coordination helps make such structures workable, while human judgement stays necessary wherever interdependencies cannot be fully specified in advance, labour relations, regulatory compliance, relationships with outside parties. [Rudko et al. \(2021\)](#) report survey evidence that most current and prospective employees hold fairly positive attitudes towards AI-driven autonomy, though this positivity is not spread evenly across a workforce. Neither paper is about tea, and that distance should stay in view.

Taken together, this literature points to a knowledge repository function, one that records the reasoning behind a grade or a price, not just the outcome. Whether participants themselves want this, and what form it should take, is a question for the interviews ahead.

2.9 What the Business Strategy Literature Offers and Where It Stops

The literature connecting AI decision support to trading and brokerage contexts specifically is limited, and the papers drawn on here operate mostly at the level of general business strategy rather than the mechanics of any particular trading platform. That distance is worth stating plainly before using them. [Vudugula et al. \(2023\)](#) argue that AI can shift organisations from reactive towards more proactive decision-making by drawing on real-time and historical data to identify patterns. This has some relevance to a tea trading platform, since surfacing historical grading comparisons alongside a new sample involves a broadly similar kind of pattern recognition, even though the underlying business and data are very different.

[Pu et al. \(2025\)](#) and [Celestin & Sujatha \(2024\)](#) report efficiency and performance gains from AI-supported decision-making in considerably larger, more data-rich organisational settings than a single-taster brokerage; their specific figures are not transferable here, but the general direction, that structured historical data used

systematically can outperform decisions made from memory alone, is consistent with the argument developed across this chapter.

This section connects back to two threads already established. The organisational literature discussed in Section 2.8 bears on platform design here too, concerning what AI does to the shape of an organisation rather than to any single decision within it. Shree Shyaam Tea Co. offers a distinctive test case for that literature's usual question, whether AI concentrates or distributes decision rights, since the firm has little existing hierarchy to redistribute in the first place; grading, pricing and buyer matching already run through one individual.

The more relevant question for a firm this size is not how to redesign a hierarchy but how to build a system that individual is willing to rely on, which returns directly to the TAM discussion in Section 2.7. Hossain et al. (2024) found that decision support systems earn adoption by demonstrably helping with the task at hand. In a business of this scale, that demonstration is unlikely to come from an aggregate efficiency statistic drawn from a larger firm; it is more likely to come from something the user can verify against their own historical grading decisions, which is a platform-design implication as much as an adoption one, and one this study is positioned to investigate directly rather than infer from literature written for a different kind of organisation.

2.10 What the Literature Prepared For — and What It Did Not

This chapter has moved through three separate literatures, and each answers a different question.

Human-in-the-loop research asks who should make the call, and it points towards AI sitting closer to comparison than to verdict (Lazaros et al. 2026, Retzlaff et al. 2024).

Explainable AI research asks how AI should support that call, and it argues the system's contribution needs to stay visible enough for an expert to question it (Chang & Bau 2025, Tchemeube et al. 2025).

TAM asks why anyone would actually use the thing, and usefulness, not polish, tends to be the answer (Davis 1989, Gefen et al. 2003, Hossain et al. 2024).

All three, in different ways, argue against building towards automated grading and towards something that makes existing judgement more consistent and visible

instead. Together, these three literatures converge on the principles summarised in Table 2.1.

The literature gives reasonable grounds to think AI can classify tea's observable and chemical traits ([Rahman et al. 2021](#), [Gill et al. 2013](#), [Yang et al. 2023](#), [Ren et al. 2023](#), [Peng et al. 2025](#)), weaker grounds to think that holds outside a controlled study ([Yang et al. 2023](#), [Gharibzahedi et al. 2022](#)), and a general case for augmentation over replacement that has rarely been tested against a single-expert trading context ([Schröder et al. 2022](#), [Ma et al. 2023](#)).

The fourth gap is where this dissertation sits: what an actual trading business needs such a system to do. [Wu et al. \(2025\)](#) note that deployed applications of deep learning in tea quality evaluation remain rare, and that interpretability and multimodal data fusion are still unresolved. [Yang et al. \(2023\)](#) trace part of this to something more basic, the absence of open, standardised tea datasets, which limits what any model can be trained or compared against. The empirical chapters that follow are built to close this fourth gap directly.

2.11 Summary

This chapter never set out to prove AI can grade tea, though evidence for physical and chemical markers points that way ([Rahman et al. 2021](#), [Gill et al. 2013](#), [Ren et al. 2023](#), [Peng et al. 2025](#)). The real pattern is Table 2.1: three sections, three principles, each preferring standardisation over automation.

Table 2.1: Literature-Derived Design Principles and Empirical Investigation Goals

Literature Findings	Design Principles	Empirical Investigation Goals
Human-in-the-loop research (Section 2.6) argues AI in judgement-heavy work functions better as comparison and reference than as automated verdict	Recommendation, not verdict: AI surfaces evidence and comparisons; the taster retains the decision	Investigate how tasters actually want to interact with AI recommendations, and where in the workflow this should sit
Explainability and TAM research (Section 2.7) shows trust depends on visible reasoning, not just accuracy, and that black-box outputs are hard to adopt regardless of performance	Explainable, visible-reasoning output: AI-generated grading support shows its reasoning rather than a bare number	Identify what level of explanation a taster needs to trust and interrogate a system's output
Knowledge management literature (Section 2.8) identifies expertise concentrated in one person as a preservation risk with no existing structured capture	Knowledge-repository function: the platform captures reasoning behind grading and pricing decisions over time	Explore how the platform can record individual expertise and support succession planning

None of this tells us what Shree Shyaam Tea Co. actually needs. That gap drives Chapter 3, which tests these principles against a single-taster brokerage, treating confirmation as validation and departure as its own finding.

Chapter 3: Methodology

3.1 *Research Philosophy*

This study adopts an interpretivist philosophy. [Saunders et al. \(2023\)](#) characterise interpretivism as concerned with the meanings people attach to their working lives, in contrast to a search for universal, law-like explanations of the kind natural science pursues. The research question seeks to understand what design features an AI-enabled platform could include to support the standardisation of tea tasting and grading in tea trading businesses, rather than testing a predetermined hypothesis about tea quality. The study therefore focuses on understanding the experiences, judgements and perspectives of the people involved in tea evaluation and trading.

This fits what the wider literature says about expertise and judgement. [Vudugula et al. \(2023\)](#) found that managers routinely fall back on intuition built from accumulated experience rather than pure logical deliberation, which is close to what happens at Shree Shyaam Tea Co., where grading decisions rest on the owner's memory and years of tasting rather than a formal rulebook. [Ma et al. \(2023\)](#) add a related, performance-conditional point: when AI accuracy is low, users are better served by an AI that helps them weigh the pros and cons of each option than by one that simply hands them a recommendation. Although neither study concerns tea grading directly, both provide support for understanding how human judgement operates before determining how AI should be designed to support it.

3.2 *Research Approach*

This study follows an inductive approach. Rather than testing a predetermined theoretical model, the research develops understanding inductively from participants' accounts of tea tasting, grading, current practices and perceptions of AI-supported decision-making, consistent with [Saunders et al. \(2023\)](#)'s view of induction as theory following from data rather than data testing theory. Existing literature, including

research on technology acceptance, human-AI collaboration and tea quality assessment, together with the preliminary discussion with the proprietor (Appendix III.2), informed the research context and the interview design, as set out in Section 3.6, but the analysis itself was not restricted to categories established in advance.

Coding follows this same logic. Themes were developed from the interview notes themselves, in the manner of open or inductive coding used by Tchemeube et al. (2025) in their evaluation of a music AI system, rather than being fixed in advance. Celestin & Sujatha (2024) similarly combine qualitative and quantitative techniques to let findings shape their conclusions, which supports this study's inductive, literature-informed stance.

3.3 Research Strategy

This research uses a two-stage qualitative design anchored in a single organisational case study of Shree Shyaam Tea Co. (see Appendix III for company background), a CTC tea brokerage in Siliguri.

Stage 1 comprised seven semi-structured individual interviews, three with focal-company participants and four with experienced professionals from other tea trading firms.

Stage 2 was a design-validation focus group, conducted after the six themes and proposed platform design had been developed from Stage 1 data. P1, E1 and E2 participated together; the proposed design was explained to them; and their collective feedback was used to assess the practicality, usefulness and acceptability of the recommendations developed in Chapter 6. Shree Shyaam was selected because its single-taster dependency and absent written protocols make grading instability and AI-adoption tensions unusually visible.

Yin (2018, cited in Saunders et al. (2023)) defines a case study as an in-depth inquiry into a topic or phenomenon within its real-life setting, noting that determining the boundaries of the case is a key factor in defining the study. This research treats Shree Shyaam Tea Co. as a single, holistic case (Saunders et al. 2023): the three focal-company participants (P1, E1, E2) sit inside the case boundary, while the four external industry professionals (IP1–IP4) are treated as comparative context from outside that boundary rather than as embedded units within it.

[Schröder et al. \(2022\)](#) used a comparable design when studying human-AI collaboration in a digital service company, arguing that a single case allows examination of a phenomenon "within its real-life context". The same logic holds here. Tea grading cannot be separated from the setting it happens in. Lighting, buyer relationships, and an owner's memory of past deals are all tied to the organisation where grading takes place.

[Saunders et al. \(2023\)](#) argue that case study strategy suits research aiming for rich understanding of a real-world phenomenon in context, particularly for 'how' or 'why' questions rather than questions of frequency. Section 3.5 explains who they were and how they were chosen.

3.4 Research Design and Data Collection

The interview instrument sits at the centre of this research design. Built around the guide approved under ERGO 115009 (Appendix I), it runs across eight sections, moving from participant background through current practice, challenges, technology use, and platform design preferences. [Saunders et al. \(2023\)](#) argue that semi-structured interviews suit exploratory and explanatory work particularly well, since they let the researcher probe and adapt in response to what a participant actually says rather than working through a fixed script. That flexibility matters here.

The structure follows an interpretivist logic: the aim is not to test hypotheses about what an AI platform ought to do, but to surface how each participant, whether proprietor, employee, or an industry contact outside the firm, understands their own tea evaluation practice and where they see room for a system to help. [Schröder et al. \(2022\)](#) used semi-structured interviews alongside other methods in a case study of AI-human task division at a digital service company, treating that combination as the way to understand how people actually work with a technology, not how a specification says they should.

Interviews took place over WhatsApp, by voice or video, since that is how this group already conducts business. They were not recorded, in line with the approved ERGO protocol. [Saunders et al. \(2023\)](#) note that where recording is likely to have a detrimental effect, it is better not to use a recorder, and that in this situation it becomes necessary to make notes instead.

The researcher accordingly took detailed written notes, using a structured Microsoft Forms template built from the sections and questions in the interview guide. Where a

follow-up call was used, participants could respond in English or Hinglish according to their own preference, matching the option set out in the Participant Information Sheet (see Appendix I). Notes remained consistent across all seven interviews while still leaving space for participants to expand on points the guide had not anticipated. Each interview ran between 30 and 45 minutes.

This design fits the interpretivist frame set out in Section 3.1. The semi-structured format gave enough consistency to compare accounts across participants, since everyone answered against the same underlying guide, while still allowing individual elaboration. One limitation is that the notes are not verbatim transcripts; summarising responses in real time carries some risk of omission or interpretive filtering.

Saunders et al. (2023) observe that note-taking styles can range from an attempt at a verbatim account to a structured record of key words and phrases, and recommend compiling a full contextual record (setting, timing, immediate impressions) alongside the notes to preserve trustworthiness where recording is not used. Using a consistent note-taking structure, tied closely to the approved guide, helped offset this, and the implications are considered further in Chapter 5. Who these participants were, and how they were selected, is addressed next.

Stage 2 of the data collection was a design-validation focus group, held after the six themes and proposed platform design had been developed (session details in Table V.2). P1, E1 and E2 participated together in a single group discussion, conducted over WhatsApp in the same manner as the Stage 1 interviews. The proposed design, as it appears in Chapter 6, was explained to them before the questions were put. Eight questions were then asked, covering perceived usefulness, workflow practicality, the human-versus-AI responsibility boundary, preferred AI functions, data storage and adoption intention. One combined note was taken; the session was not recorded, consistent with the approved ERGO protocol.

Because the discussion was collective, responses cannot be attributed to individual participants. The focus group was not treated as a fourth or eighth participant. It did not generate themes. Its function was to validate whether the design recommendations were practical and acceptable to the people who would use the system.

3.5 Sampling Strategy and Participant Profile

Participants were selected through purposive sampling, following the case study design set out in Section 3.3. The logic here is simple: tea grading is an expert skill, built over years of practice, not something any adult can do reliably. So the sample had to be built around people who already hold that skill, rather than drawn from a broader, representative population. [Bhattacharyya et al. \(2008\)](#) describe tea tasting as requiring "ultrasensitive classification ability", which points to just how specialist the task is. [Peng et al. \(2025\)](#) report that professional tasters typically need more than five years of training. That finding shaped the study's own criterion: participants needed a minimum of five years' experience in tea tasting and grading, and had to be adults working within the Indian tea industry.

This follows [Saunders et al. \(2023\)](#), who characterise purposive sampling as selecting participants deliberately for their relevant knowledge or characteristics, rather than through random selection aimed at statistical representativeness.

The Stage 1 interview sample comprised seven participants: three from within the case boundary (the proprietor of Shree Shyaam Tea Co. and two employees working directly under him), and four industry peers of comparable seniority from separate trading businesses, positioned outside the case boundary and included for the cross-company comparison described in Section 3.3, consistent with Yin's (2018) distinction between the case itself and external comparators ([Saunders et al. 2023](#)).

The Stage 2 focus group involved the three focal-company participants, P1, E1 and E2, who had already taken part individually in Stage 1. No new participant was introduced. The unique participant count remains seven. A planned sample of five to ten participants suited the focused nature of the study and the specialist nature of the population. Comparable qualitative work has shown that small expert samples, drawn from bounded organisational contexts, can still generate real depth of insight ([Schröder et al. 2022](#)). Seven participants gave perspectives from both the focal organisation and the wider trading context. Recruitment ran through the proprietor as gatekeeper (Appendix I), whose standing across the regional trading network made direct, trusted introductions possible in a way cold recruitment never could have been.

3.6 Instrument Design and Preliminary Development

The interview guide grew out of a preliminary discussion with the proprietor of Shree Shyaam Tea Co. (summarised in Appendix III.2), held while the research proposal was still being developed. That conversation surfaced the problems that shaped the research question: grading that rests on personal judgement, lighting and weather interfering with visual assessment, and the difficulty of holding onto tasting expertise as people move on. It shaped the topics the guide covers. It was not used to test wording, sequencing or duration, so it is treated here as a preliminary discussion rather than a pilot.

Questions were then tied back to specific parts of Chapter 2 rather than invented from scratch. Table IV.1 (Appendix IV) sets out that mapping in full, showing each literature finding, the sources behind it, the section it comes from, the initial code attached to it, and the questions it produced. Items on grading inconsistency and taste memory follow from the sensory-evaluation literature in Section 2.2. Items on AI use, trust and platform features draw on the acceptance literature, particularly Davis (1989) and Sohn & Kwon (2020), so that participants responded to a shared reference point rather than whatever they happened to picture on hearing the word "AI." Three questions carry no literature anchor and came instead from the case context.

The Stage 2 focus-group guide was a separate instrument with a different purpose: eight questions assessed the proposed platform design rather than tea evaluation practice, covering usefulness, workflow benefit, tray-system practicality, the human-AI responsibility split, AI-function comfort, data security, and adoption intention (full wording and responses in Appendix VII, Table VII.1). The questions did not replicate the exploratory interview instrument and were not coded into the six-theme framework; the focus group instead validated the Chapter 6 recommendations against prospective users' experience.

3.7 Data Analysis

Interview notes were coded systematically and iteratively rather than sorted into categories fixed in advance. Saunders et al. (2023) describe TA as identifying patterns of meaning across a dataset, moving from initial coding to broader themes in a process that stays iterative and works across different epistemological positions. The six phases followed here, familiarisation, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final

analysis, follow [Braun & Clarke \(2021\)](#), whose reflexive approach treats the researcher's judgement as part of the analysis rather than a source of bias to be removed. This fits the interpretivist stance taken in [Section 3.1](#) and the inductive approach in [Section 3.2](#).

Initial codes were seeded from the [Chapter 2](#) themes, set out in [Appendix IV](#), the same [appendix Section 3.6](#) uses to map interview questions to their literature sources, with room left for codes the literature had not anticipated. Coding was not linear; early codes were revised as later notes were reviewed. Where a participant's account sat in tension with the wider pattern, it was retained and reported as a point of divergence. The interviews were not recorded, so coding worked from detailed notes taken during each WhatsApp call. How many initial codes were generated, and how they collapsed into sub-themes and final themes, is reported in [Chapter 4](#).

Stage 1 interview data were analysed using the iterative thematic analysis described above. The Stage 2 focus group was handled separately. Its responses were not entered into the six-theme coding matrix, because the focus group took place after those themes and the proposed design had already been developed. Instead, the combined response was mapped against the proposed platform features to identify where participants confirmed the design, where they suggested refinement, and whether any additional requirements emerged. This is reported in [Chapter 6](#), consistent with member validation's function as confirmation of prior findings rather than as new analysis.

3.8 Trustworthiness

Given the interpretivist and qualitative design, trustworthiness was considered through Lincoln and Guba (1985, cited in [Saunders et al. \(2023\)](#)) parallel quality criteria of credibility, transferability, dependability and confirmability, developed as qualitative equivalents to positivist reliability and validity. Each one needed an actual tactic, not just a name-check.

For confirmability, the analysis was checked reflexively against the raw interview notes at each stage of coding, tracking where an emerging theme was, and was not, directly supported by what participants said, rather than by the researcher's own expectations. Dependability came from sticking to the same interview guide and note-taking structure across all seven participants. Given the single-case design, statistical generalisation is not the objective; instead, detailed description of the

organisational and industry context allows readers to assess the potential transferability of the findings to comparable tea trading settings. These tactics are summarised in Table 3.1.

Table 3.1: Trustworthiness criteria and tactics used

Criterion	Tactic used
Credibility	Comparison of perspectives across proprietor, employees, and external industry participants in Stage 1, followed by collective design validation within the focal organisation in Stage 2
Transferability	Thick description of company and industry context
Dependability	Consistent interview guide and structured note-taking process across Stage 1, and a structured eight-question focus-group guide for Stage 2
Confirmability	Reflexive audit trail checking emerging themes against raw interview notes; Stage 2 focus-group feedback kept separate from thematic coding and used only to validate the Chapter 6 design recommendations

For confirmability, the analysis was checked reflexively against the raw interview notes at each stage of coding, tracking where an emerging theme was, and was not, directly supported by what participants said, rather than by the researcher's own expectations. The preliminary discussion that shaped the interview guide (Appendix III.2) was treated as a separate input to instrument design rather than as interview data, and was not itself coded as a source of findings.

3.9 Ethical Considerations

This study received ethical approval from the University of Southampton (ERGO 115009). All participants were provided with a Participant Information Sheet and Consent Form prior to interview, and participation was entirely voluntary, with consent obtained in writing beforehand. No audio or video recording took place; data collection followed the approach described in Section 3.4.

The Stage 2 focus group was conducted under the same ethical arrangements, as confirmed by the supervisor as an extension of the approved interview process under ERGO 115009; no separate amendment was required.

Participants were assigned identification codes, and identifying information was separated from the interview data wherever possible. Data were stored securely on University-approved OneDrive, destroyed after conferment of the degree. Participants could withdraw within 14 days of interview, after which withdrawal would no longer remove their data. Full documentation is provided in Appendix I.

3.10 Reflexivity

Shree Shyaam Tea Co. is my father's business. I grew up with some family knowledge of the trade but had no operational role in it before this research, and my interest was partly personal, tied to my own curiosity about tea as a start-up area. Although I did not enter the interviews with a predetermined view of which AI features participants should support, the proposal-stage discussions and existing literature had already shaped my awareness of potential applications, so I cannot claim to have approached the topic without any prior framing.

Most participants already knew me through family ties, which gave fast access but carried risk: familiarity can make a researcher hear confirmation where none exists. This familiarity extended to the preliminary discussion that shaped the research direction and interview guide before data collection began (Appendix III.2), where my existing relationship with the proprietor gave fast, trusting access but also meant the conversation was not held at arm's length in the way an interview with a stranger would have been. Chang & Bau (2025) note that analysis drawn from a researcher's own experience can carry personal bias even when the method looks systematic. To manage this, I treated participants' accounts as the primary basis for theme development and retained views that challenged the assumptions identified during the proposal stage, rather than filtering the notes towards what I expected to find.

A further reflexivity consideration applies to the Stage 2 focus group. P1 is the proprietor; E1 and E2 are his employees. In a group setting, that power dynamic may have made the employees less likely to openly challenge aspects of the proposed design, even where reservations existed. The combined note, and the absence of individual attribution, mean there is no way to recover whether dissent was suppressed or simply absent. Stage 2 responses are therefore treated throughout as

collective case-company feedback rather than three independent assessments.

3.11 Summary

This chapter has laid out an interpretivist, literature-informed, single-case design comprising seven semi-structured individual interviews across the focal company and four industry peers, followed by a case-company focus group used to validate the proposed design recommendations, with all interview data analysed through an iterative, data-driven thematic analysis. The next chapter presents what that process actually surfaced, moving from method to findings.

Chapter 4: Findings

4.1 Overview and Participant Profiles

Seven people took part in this study, drawn from three groups connected to Shree Shyaam Tea Co. and the wider Indian tea trade: the business owner (P1), two employees (E1, E2), and four industry peers split by tenure into a younger pair (IP1, IP2) and a more experienced pair (IP3, IP4). Selection was purposive. Tea grading is a skill built over years, not something any adult can reliably do, so the sample had to be built around people who already hold that skill. Every participant had at least five years' experience tasting and grading tea, and all worked within the Indian tea industry as adults.

Roles varied more than the group labels imply. P1 covers the most ground of anyone interviewed, tasting, buying, price negotiation, and buyer relationships all sit with them. IP3 and IP4 cover similarly broad territory; IP4 also handles auction bidding and blend approval for packet orders. At the other end, E1 works mainly on negotiation and buyer relationships rather than tasting directly, and E2's role is narrower still, focused on day-to-day tasting and grading. Tenure ranges from eleven years (E1, E2) to over twenty (IP3), so even the "experienced" band is not a uniform group. Job title alone does not predict how much ground someone covers either, P1 and IP4 both range across the full tasting-to-negotiation chain despite sitting in different stakeholder groups.

Interview procedure follows Section 3.4. A full breakdown of each participant, role, tenure, and relevant background, is provided in Table V.1 (Appendix V) . The themes discussed in the sections that follow draw on the coded evidence set out in full in Appendix VI.

4.2 Empirical Findings on Tea Grading, AI and Platform Requirements

Theme 1: Grading Is Experiential and Context-Dependent, Not Standardised

The title of this theme is only half true, which is what makes it worth keeping. Grading turns out to hinge less on personal instinct in general and more on which side of the case company's walls a person sits. Inside Shree Shyaam Tea Co., a grade label reads as fixed. Outside it, the same label reads as a starting point that still needs a human eye to finish the job.

The split shows up almost word for word (see Figure 4.1). P1, E1, and E2 each independently called grading "a fixed industry standard," all three using near-identical phrasing. All four external peers said close to the opposite. IP1, IP2, and IP4 all gave the identical answer: I mainly rely on my own judgement, regardless of the label." IP3 went further, describing grade meaning as something that varies significantly by garden or region."

A plausible reading: inside one company, a label like BOP or PF works because everyone in the room has already calibrated to the same reference points, tasted the same lots, answered to the same buyers. It feels fixed because the calibration happened once, collectively, some time ago. Traders working across many counterparties never get that shared reference point. For them the label is an assumption, and personal judgement fills the rest.

P1's own data does not sit still, though. In the same interview where P1 calls grading fixed, P1 also notes that aroma expectations shift what the same label means between gardens, without seeming to notice the two claims do not fit together. That single contradiction is more revealing than the clean company-versus-peer split it lives inside, and it is retained in the coding matrix rather than smoothed over.

The divide does not track experience cleanly either. Split the sample by tenure instead of company, and the fifteen-plus-year band (P1, IP3, IP4) splits one fixed to two individual, while the under-fifteen-year band (E1, E2, IP1, IP2) splits evenly, two and two. Neither band lines up with company membership the way the raw split does. So this is not about how long someone has spent tasting tea. it is about who they taste it for.

That matters for the study's actual question. A model trained on Shree Shyaam's internal grading conventions is learning one company's shared calibration, not the

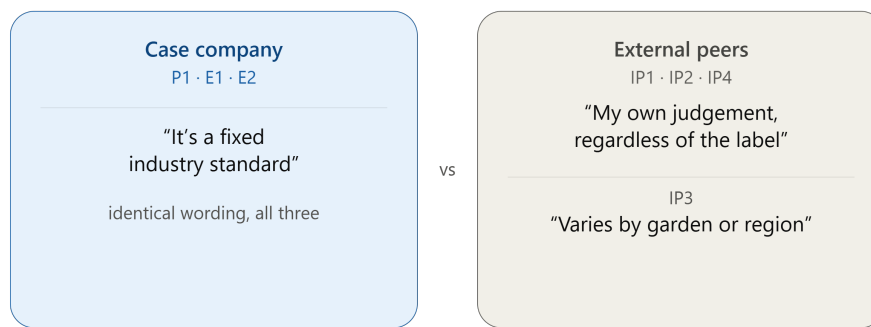


Figure 4.1: The fixed-standard versus individual-judgement split: all three case-company participants describe grading as a fixed industry standard, while three of the four external peers rely on personal judgement and the fourth describes grade meaning as varying by garden or region.

trade's. Put it in front of outside buyers or auction counterparties who read the same label differently, and it would be exporting a local agreement as a universal standard, the exact gap this theme's title, read too literally, would paper over. The more honest version of the finding is not "grading is unstable"; it is what counts as fixed depends on which side of the company boundary you are standing on.

Theme 2: Trust in AI Is Conditional and Evidence-Based

Participants did not split into "for" and "against" AI so much as into different lists of proof they would want first. The theme is less about optimism or scepticism and more about what evidence each person needs before they could hand over a grading call.

That conditionality took two forms. Some wanted a one-time check, an explanation, a side-by-side comparison against a sample they already knew. Others wanted something closer to a track record, trust built up over repeated use rather than granted on the first right answer.

P1 stacked several conditions: an explanation for the output, a comparison against a past sample, and the ability to override it. E1 put it more simply, trust requires "the ability to review and override the AI's output." IP1 wanted "accuracy proven over time" rather than an explanation alone, and IP3 proposed testing a system "beside with me for a month or so" before trusting its calls. E2 sits at the lighter end of the group, the most enthusiastic participant about AI overall, but the one naming only a single condition (a clear reason for the grade given).

IP4 breaks the pattern. "No camera has either", meaning taste or smell, is not a bar more evidence could clear. it is a claim about what the technology cannot do, full stop, not a threshold it has not reached yet. Filing that under "conditional trust" undersells it.

The cross-group data points to experience, not company membership, as what drives this. All three participants with 15+ years (P1, IP3, IP4) stack multiple conditions and name several tasks that should stay human-only. The four under 15 years (E1, E2, IP1, IP2) each name just one condition and one human-only task, regardless of whether they work inside the case company or outside it (Figure 4.2). Worth flagging: the "fixed standard vs. individual judgement" split from Theme 1 runs along company lines, and this one does not. Two different variables, doing two different jobs in the data.

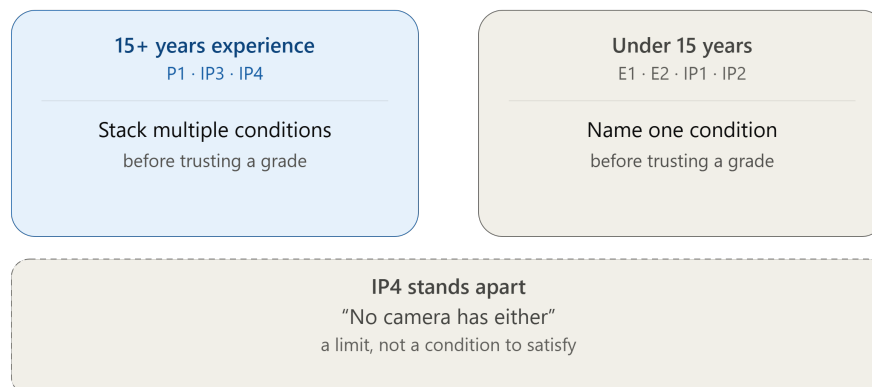


Figure 4.2: Trust conditions grouped by tenure rather than company: the three participants with 15+ years' experience stack multiple conditions before trusting an AI grade, while the four under 15 years each name only one. IP4, though in the experienced group, holds a categorical objection rather than a conditional one.

IP4's refusal on taste and smell deserves to stand apart from the rest of this theme rather than get folded into it. Everyone else is asking AI to earn trust. IP4 is saying there is a part of the job it was never going to be able to earn trust on in the first place.

Theme 3: AI's Realistic Value Lies in Support Functions, Not Sensory Judgement

The coded evidence for five participants (E1, IP1, IP2, IP3, IP4) lines up behind this framing: AI belongs in the record-keeping and decision-support layer, not in the sensory call itself. P1 and E2 sit slightly closer to that line, but not by rejecting it (Figure 4.3).

The split works like this. On one side, there is genuine appetite for AI as an operational aid: record-keeping, matching lots to buyers, price guidance, drafting tasting notes, holding on to institutional knowledge before it walks out the door with a retiring taster. IP2 names the widest list of these: records, lot matching, price guidance, notes, decision support. IP1 cares mostly about continuity, hoping AI can help "preserve tea-tasting knowledge for the future."

On the other side sits a line nobody in this group wants crossed. IP4 states it flatly: "Half of what I judge never reaches the eye." That is not a complaint about current technology rather it is a claim that no camera will ever fix, because the thing being judged was never visual to begin with. IP4 accepts AI for history and record-keeping, full stop, nothing to do with grading. IP3's version is gentler but points the same direction: a system that "sits with" trainees and shows them comparisons, not one that grades in their place. Training aid, not replacement grader.

The study's two most AI-positive voices (P1 and E2), are also the ones who lean closest to the sensory line. P1's wish list includes image analysis and lighting correction sitting right alongside buyer matching and price guidance, functions that are harder to call purely administrative. E2's optimism runs through a commercial hope: that AI could cut down disputes with buyers, which quietly requires trusting a grading-adjacent output, even if E2 never says outright that AI should taste. Worth noting too: the two owners in the sample, P1 and IP3, are also the two asking for the broadest AI feature sets overall, a pattern that tracks ownership, not which side of the company boundary someone sits on.

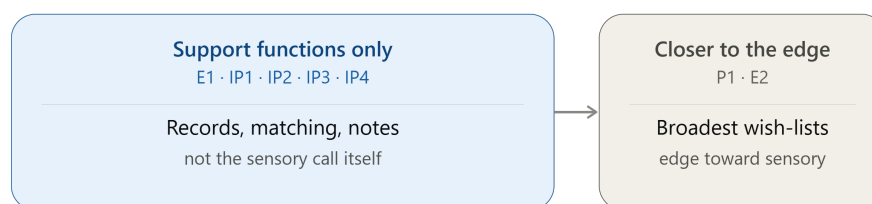


Figure 4.3: AI's role as a spectrum rather than a hard split: five participants confine AI strictly to support functions, while P1 and E2 sit nearer the sensory edge without actually asking AI to taste or smell anything.

So the theme holds, but it is not unanimous in the same way for everyone. Five participants treat "support, not sensory judgement" as close to a hard rule. P1 and E2 stand nearer the edge of it. One pushed there by the demands of running a business, the other by wanting fewer arguments with buyers, but neither actually asks AI to taste or smell anything. What both groups agree on, underneath the

disagreement about scope, is that judgement itself stays human. That is the part worth carrying into the next theme.

Theme 4: Human Judgement and Accountability Must Remain Central

Nobody in this study was willing to let an AI grade stand on its own. Every participant, without exception, drew a line around final authority. The only theme with zero individual-level dissent, though how people planned to keep control differed more than expected.

P1 and E2 described silent override: if the AI disagreed with them, they would just overrule it and carry on, no conversation needed. E1, IP1, IP2, and IP3 all converged on a different answer to the same question, a system that "learns from my correction", treating disagreement as training signal rather than a one-time veto. IP4 alone wanted something more formal: "flag it for review," rather than letting a correction pass silently.

IP3's line stuck with me: "The final call must stay with the person who has to answer to the buyer." That is not really a statement about whether the AI is accurate. Even a system IP3 fully trusted would still need their signature, because they are the one who has to explain a bad sale to a buyer, not the software.

The split here does not actually track the case-company/external line the way the rest of the data does. Silent override is P1 and E2 only, two of the three case-company participants. Everyone else, five of seven, lands on "learns from my correction," including E1, who works inside the same company as P1 and E2. Only IP4 wants a formal flag-for-review step. So the real contrast is P1/E2's unilateral correction against a five-person majority favouring an ongoing, coaching-style relationship with the tool, company membership does not predict it (Figure 4.4). What holds across every group is the commitment itself: nobody, anywhere in the sample, would let a grade go through unchecked.

I would resist treating "human judgement stays central" as one clean finding, because it is not really one mechanism. it is at least three, and they do not behave the same way. Silent override and formal flag-for-review are tight: the system's behaviour does not drift between one grading session and the next, because a human catches every instance. "Learns from my correction" is looser by design because between corrections, the tool is running on whatever it last absorbed, and that is a real gap the other two approaches do not have. Grouping all three together is not wrong, exactly,



Figure 4.4: Three ways of keeping a human in control of an AI grade. "Learns from my correction" is the majority position (E1, IP1, IP2, IP3); silent override (P1, E2) and a formal flag for review (IP4 alone) are the minority positions. All seven participants keep a human in final control, but the mechanism differs.

but it hides a distinction that matters if anyone actually tries to build and audit one of these systems.

The honest version of this theme, then, is not "people do not trust AI to grade alone." it is that everyone agrees the human stays accountable, but nobody has actually agreed on how loose the leash gets in between, and that is the part a real deployment would have to sort out.

Theme 5: Data Control and Cost Sensitivity Shape Adoption Willingness

Every participant had a clear view on where their data should sit and what they would pay for a tool. The two do not track each other. Being cautious about data does not predict being cautious about cost.

Storage leaned local. P1, IP1, IP2, and IP4 all said local only. E1 and IP3 both said it depends what is stored. E2 was the only one comfortable with cloud outright. Even "local" meant different things to different people: IP2, IP3, and IP4 pointed to a specific worry, competitors getting at their records, while IP1 gave a general privacy instinct rather than anything tied to competition. P1 broke from the group entirely, local storage, but "no particular concern" behind it.

Cost is where the spread showed up. E2 wanted the tool free. P1, IP3, and IP4 would each stretch to a small one-time cost, equipment-level, nothing recurring. E1 and IP2 said they would invest significantly, but only once the tool proved itself. IP1 landed in between, fine with a small monthly subscription.

Line the two up and the pairing breaks twice, not once. IP1 is privacy-cautious but

cost-open by choosing a recurring subscription and no proof demanded first. IP2 is privacy-cautious for a named reason and still willing to pay significantly, once the tool earns it. Neither fits the assumption that data caution and cost caution travel together. IP2 and IP3 both cite the same worry, competitor access, and still end up at opposite ends of the cost scale from each other.

The case company does not hold together internally either (Figure 4.5). E2's free-cloud stance and E1's depends-on-content-but-invest-once-proven stance sit almost as far apart as two answers can get, despite coming from the same business. P1 pairs local storage with a low cost tolerance, in the same small-one-time-cost bracket as IP3 and IP4, though with barely any stated reasoning behind either choice. E2 sets the real floor on cost, wanting the tool free outright.

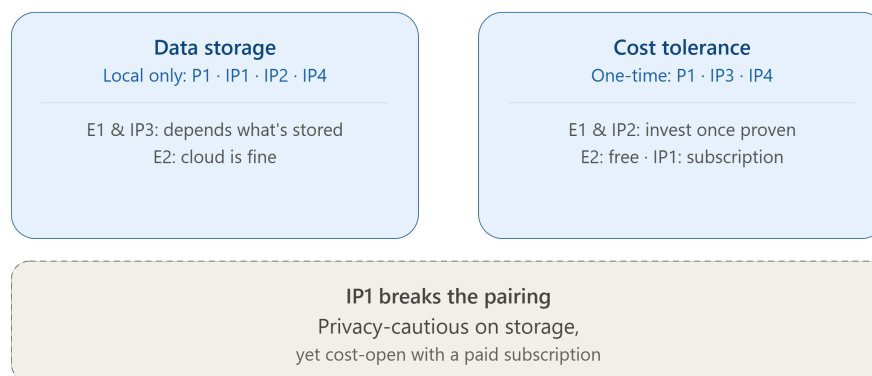


Figure 4.5: Data storage preference and cost tolerance as two independent variables. IP1 is the clearest break: privacy-cautious on storage, yet the most cost-open participant, accepting a recurring subscription with no proof demanded first.

All seven participants gave a clear position on both storage and cost. What varies is the combination, and that variation does not sort neatly by company, by role, or even by what specific worry someone named. A local, pay-once tool would suit P1 and IP4 well. A free cloud tool suits exactly one person, E2, and nobody else. Anything built for this market has to treat data sovereignty and pricing as two separate decisions, because the seven people in this study clearly do.

Theme 6: The Deeper Business Problems Lie Beyond Grading Itself

Grading disagreements were rarely the thing participants actually worried about. What kept coming up instead was knowledge that never gets written down, records scattered across formats no one else can read, and a sense that the real risk sits

somewhere else entirely.

Two things drove this. First, succession. P1 wants AI to capture "general experience/intuition, hard to put into words," so the business is not stuck depending on one person. IP3 said it more bluntly: "I have people with five, six years who still cannot tell me why they gave a grade." Both are owners. Both are describing the same fear from opposite sides: one worried about P1's own knowledge disappearing with them, the other about staff who taste correctly but cannot say why.

Second, the record-keeping is a genuine mess. P1 works off personal notes and paper. E1 uses Excel and nothing else. IP2 admitted buyer preferences are tracked with "no formal system". Only IP1 had gone beyond spreadsheets, running Excel plus a separate digital system. E2's addition here does not quite fit either bucket: blending came up in the interview conversation, not in answer to a specific question about it. It is a sign of what tends to get left out of the structured parts of these conversations altogether.

IP4 is the one who complicates things. The real problem is not grading. it is auction samples not matching what actually arrives in the chest, which he called "a bigger loss to me than any grading disagreement." But ask him about capturing tacit, hard-to-verbalise knowledge and he does not select it. That is not a small omission. Tacit knowledge only ever came up from the two business owners, P1 and IP3. Nobody else, not the three employees, not the other two industry peers.



Figure 4.6: Tacit or hard-to-verbalise knowledge is named only by the two business owners, P1 and IP3. IP4, despite using the most vividly sensory language in the study, does not select it when asked directly.

(Figure 4.6) It tracks with something else in the data: the case company runs almost entirely on Excel, paper, and personal notes, while the outside peers use a scrappier mix: diaries, phone photos, even WhatsApp for sending buyers sample images.

There is a real theme here, but it is not quite the one it looks like at first glance.

Everyone agrees grading is not the deep problem. They just do not agree on what is. Owners keep circling back to knowledge walking out the door. Everyone else is worried about auction integrity, informal tracking, or something they have not quite named yet. Treating that as one shared concern would erase a difference that is genuinely worth keeping. The people carrying the commercial risk are worried about a different thing than everyone around them.

4.3 Summary

Across the seven interviews, six themes emerged, and the coded evidence underpinning each is set out in full in Appendix VI. Theme 1 (Table VI.1) shows grading read as fixed within the case company but individually negotiated outside it, with P1's own account holding both positions at once. Theme 2 (Table VI.2) traces trust in AI as something built through stacked conditions, evidence, comparison, override, except for IP4, whose objection is categorical rather than a bar more evidence could clear. Theme 3 (Table VI.3) confines AI's realistic role to support functions, with P1 and E2's broader wish-lists the only entries that edge towards sensory territory.

Theme 4 (Table VI.4) is the one theme with no coded contradictions: every participant keeps a human in final control, though the mechanism, silent override, correction-based learning, or formal flagging, varies. Theme 5 (Table VI.5) shows data caution and cost tolerance moving independently rather than together, IP1 being the clearest break in that pairing. Theme 6 (Table VI.6) locates the deeper business risk in tacit knowledge and record-keeping rather than grading itself, though only the two owners, P1 and IP3, flag tacit knowledge directly, IP4's evidence contradicting that particular sub-thread while supporting the theme's broader claim.

Chapter 5 reads these six findings against the literature reviewed in Chapter 2.

Chapter 5: Discussion

5.1 Introduction

The study asked what an AI-enabled platform would actually need to do to support the standardisation of tea tasting and grading inside a small Indian brokerage. Chapter 4 answered that through seven interviews. This chapter puts those answers in conversation with the literature that shaped the research questions.

That conversation does not go smoothly in every case, and it would be dishonest to pretend otherwise. Some findings land where the literature expected them to. Theme 2's conditional trust conditions map almost directly onto the explainability and TAM literature in Section 2.7; Theme 4's human-in-the-loop consensus sits squarely within Section 2.6.

Others land close but with complications the prior work did not prepare for: Theme 1 complicates rather than confirms Section 2.2's subjectivity claims, because inconsistency turns out to track organisational boundaries rather than individual variation. One or two land somewhere the literature does not really reach: the data sovereignty and cost findings in Theme 5 sit only partially within the SME adoption literature, and the tacit knowledge concerns in Theme 6 belong to owners in a way the knowledge management literature treats as incidental.

Section 5.2 sets out that mapping in outline, Section 5.3 works through each theme's fit with the literature in turn, and Section 5.4 draws the six threads into what the platform must do.

5.2 How the Themes Are Mapped to the Literature

Six themes emerged from the analysis, each tracing back to a different part of Chapter 2. Table 5.1 sets this out in advance, so the logic behind the ordering is

visible from the start.

Each row holds four things: the finding, its underlying code clusters (from Appendix IV and Appendix VI), the design principle or investigation goal it tests, and its Chapter 2 anchor. The final column states what the finding would ask a platform to do differently.

The rows do not carry equal weight. Themes 2, 4 and 6 test the three design principles in Table 2.1 fairly directly. Themes 1, 3 and 5 sit closer to the investigation goals, and some codes beneath them, specifically data locality, privacy reasoning and assessment tempo, arose from the case context rather than the reading. The cost and resource side of Theme 5 remains anchored in Sections 2.8 and 2.9. That asymmetry matters: it marks where the literature spoke to a small brokerage and where it did not.

Table 5.1: Synthesis of empirical findings, literature connections and design implications

Finding theme	Correlating interview codes	Design principle / investigation goal	Literature section	Design implication
T1. Grading is experiential and context-dependent, not standardised	SUBJ- BASIS, SUBJ- DRIFT ENV- LIGHT, ENV- WORKAROUND FAULT- VETO, TIME- PRESSURE	Goal: establish how far grading varies and what drives that variation, testing the premise all three principles rest on	2.2, 2.3 (framing in 2.1)	Comparison has to be anchored to garden, season and light conditions, not to the grade label alone. A calibration set built inside one firm cannot be shipped as an industry standard
T2. Trust in AI is conditional and evidence-based	EXPLAIN- NEED, TRUST- EVIDENCE AI- SCEPTICISM TAM- EASE	Principle 2, explainable, visible- reasoning output. Goal: establish what level of explanation earns trust	2.7	Every output carries a stated reason and a named past sample beside it. A shadow or trial period before live use, since some participants want a track record rather than an explanation

Continued on next page

Table 5.1 continued from previous page

Finding theme	Correlating interview codes	Design principle / investigation goal	Literature section	Design implication
T3. AI's realistic value lies in support functions, not sensory judgement	CV- VISIBLE, CV- BLINDSPOT AI- ROLE SENSE- TRADEOFF	Goal: locate where in the workflow AI support belongs, and where it does not	2.4, 2.5, 2.6	Feature scope stops at what a camera and a record can actually hold. No aroma or taste score is generated, including as a suggestion the taster could lean on
T4. Human judgement and accountability must remain central	OVERRIDE HUMAN- RESERVED AI-ROLE	Principle 1, recommendation rather than verdict, with the human retaining final authority	2.6	Override behaviour configurable across the three mechanisms found in the data: silent correction, learning from correction, and flag for review. Each change logged against a named taster
T5. Data control and cost sensitivity shape adoption willingness	DATA- LOCALITY, PRIVACY- CONCERN COST- THRESHOLD, RESOURCE- LIMIT TAM- USEFUL	Goal: identify the resource and governance conditions adoption depends on. Partly case-context, anchored in the SME constraint literature in 2.8 and 2.9	2.8, 2.9 (case-context Appendix IV)	Local- first storage with a one- time cost over. Storage and pricing treated as two separate decisions, since the two move independently across the sample

Continued on next page

Table 5.1 continued from previous page

Finding theme	Correlating interview codes	Design principle / investigation goal	Literature Design implication section
T6. The deeper business problems lie beyond grading itself	TACIT- KNOWLEDGE, SUCCESSION RECORD- PRACTICE, ANCHOR- SAMPLE HIST- COMPARISON, BUYER- MATCH PRICE- JUDGEMENT	Principle 3, knowledge-repository function. Goal: explore how individual expertise is recorded and passed on	2.8, 2.9 The record holds the reasoning behind a grade, not just the grade, together with who tasted the sample and when. Buyer preference stored as data rather than recall

Note. Code clusters are shortened here for readability; the full set of codes and the questions they came from is in Appendix IV, and the coded evidence behind each theme is in Appendix VI. Design principles and investigation goals are those set out in Table 2.1.

5.3 Where the Findings Land and Where They Do Not

What follows takes each theme in turn and presses it against the literature directly. Each subsection moves from the finding, to what the literature said to expect, to what that leaves the platform needing to do; where the fit is clean, it is stated as such, and where it is not, the friction is the finding.

Where Grading Instability Actually Comes From

The literature framed grading inconsistency as a property of individual tasters. This study found something more specific. P1, E1, and E2 each independently called grading "a fixed industry standard," near-identical phrasing across three separate interviews. Every external peer said roughly the opposite, relying on personal judgement regardless of the label, with IP3 describing grade meaning as varying by garden and region. Splitting the sample by tenure produces no pattern. The variable that matters is organisational membership. P1's own account makes this stranger still: in the same interview, P1 also notes aroma expectations shift what the same label means between gardens, apparently without noticing the two claims contradict each other.

That organisational pattern is not what the existing literature was built to explain. [Bhattacharyya et al. \(2008\)](#) and [Laddi et al. \(2012\)](#) treat subjectivity as a property of the individual taster: fatigue, prior exposure, state of mind. [Sajeewani et al. \(2023\)](#) makes a related but narrower claim that flavanol composition correlates with taster scores closely enough to replace the taster with a trained model.

Section [2.3](#) adds environmental noise from lighting and seasonal chemistry. Both framings assume instability scatters around a shared signal, one that better conditions or training could tighten. This data does not fit that picture. What the literature calls individual variation may, in some settings, be calibration disagreement between firms working from different shared references. That is a different problem structurally, and the environmental literature in Section [2.3](#) does not reach it.

A platform trained on one firm's grading history learns that firm's internal agreement about what BOP or PF means, not the trade's. Presented to buyers or counterparties working from a different shared reference, it exports a local calibration as an industry standard and reproduces the disagreement it was meant to reduce. The design response is not better lighting controls. It is a system that knows its calibration is local and says so.

Conditional Trust: Explainability and the Limits of Evidence

Most participants would not accept an AI grade without conditions first. P1 stacked three before trusting any output: an explanation, a comparison against a past sample they already knew, and an override right. IP3 wanted a month of parallel running before going live. E2 asked only for a clear reason behind the grade. IP4 is a different case entirely. "Taste is in the mouth and aroma is in the nose, and no camera has either." That is not a bar set high. It is a door closed.

[Gefen et al. \(2003\)](#) argued that trust shapes whether perceived usefulness translates into actual use. [Corvelo Benz & Gomez Rodriguez \(2023\)](#) add precision the earlier TAM literature lacked: confidence values alone do not build trust; what matters is whether the system's stated confidence aligns with the user's own. P1 and IP3 behave exactly this way, using explanation requests and trial periods to actively calibrate the AI against their own expertise. [Chang & Bau \(2025\)](#) define explainability as what allows a user to understand how an output was reached. The data validates this for six of seven participants. IP4 sits outside it. Explainability frameworks assume trust is a threshold visibility can eventually reach. IP4's objection is not about threshold. It is about category.

This has a direct consequence for platform scope. A system built to earn trust through explanation and comparison will work for most people in this sample. It will not work for practitioners who hold what is best described as an ontological objection to machine sensory judgement, not scepticism about current accuracy, but a view that tasting and classifying are not substitutable at all. The explainability literature does not model this possibility. IP4's position is a conceptual edge case for [Chang & Bau \(2025\)](#). For feature decisions, it functions as a hard constraint.

Support, Not Sensory Judgement

Five participants drew the same boundary without coordinating on it. E1, IP1, IP2, IP3, and IP4 all confined AI to records, lot matching, price history, and tasting notes. Nobody wanted a system that grades. IP4 put it plainest: "Half of what I judge never reaches the eye." That is not a complaint about current accuracy. It is a claim about what grading is. The nose and palate carry most of the load; a camera reaches neither. P1 and E2 push closer to the sensory edge, with wish-lists that include image analysis, but even they stop short of asking AI to taste or smell anything.

[Zhi et al. \(2025\)](#) state directly that computer vision reads only surface attributes and cannot capture the chemistry that determines a tea's value in the cup. [Gharibzahedi et al. \(2022\)](#) make the same point about multi-sensor arrays: no single instrument holds aroma, taste, and appearance together. Sections 2.4 and 2.5 drew on both papers to argue the design question should shift from what AI can measure to where AI and taster should work together. The empirical data confirms that shift, from inside the industry. IP4's "half of what I judge never reaches the eye" is close to a practitioner restatement of [Zhi et al. \(2025\)](#) camera-versus-chemistry point, arrived at independently. The [Schröder et al. \(2022\)](#) augmentation framing from Section 2.6 sits behind this too: every participant described augmentation, not one described sensory automation.

This theme rules out a category of feature before anything is built. No aroma score, no taste prediction, not even as a suggestion the taster could override. The reason is not modest ambition. If the object of judgement is partly invisible to a camera, a score anchored to surface attributes alone is a number built on incomplete evidence, and a taster who leans on it does not know what they are missing. Ruling out sensory AI leaves a substantial feature set intact: records, comparisons, buyer matching, price guidance, stored notes. Confining the platform to support functions is a design decision grounded in what grading actually is, not a concession to what the

technology cannot yet do.

Human Accountability and the Mechanisms of Oversight

Seven participants, zero dissent: the final call stays human. But that consensus hides the real finding. P1 and E2 described silent override, overruling the system and moving on, nothing logged. E1, IP1, IP2, and IP3 wanted correction-based learning, a system that absorbs disagreement as training rather than absorbing a one-time veto. IP4 alone wanted formal flagging, a structured pause before anything proceeded. The principle is unanimous. The mechanism is not, and that difference matters more than the headline agreement suggests.

Design Principle 1 asks for recommendation rather than verdict, and [Lazaros et al. \(2026\)](#) and [Retzlaff et al. \(2024\)](#) both frame human presence as the safeguard against AI substituting for expert judgement. The data confirms the principle but exposes a gap in the theory. Human-in-the-loop literature treats presence as binary: either the person retains authority or they do not. It says almost nothing about what happens between override moments. "Learns from correction" is the majority position here, and it is also the loosest mechanism. Between corrections, the system runs on whatever it last absorbed, a drift [Ma et al. \(2023\)](#) did not quite anticipate.

Configurable override behaviour is not optional. A platform built around one mechanism would leave P1 and E2 working around it, which is an adoption failure when the case company's proprietor is a primary user. There is also an audit problem the literature has not caught up with. Silent override leaves almost no trace: the grade changed, nothing logged. Correction-based learning generates the richer record, even though the system has more latitude between corrections. The group producing the tightest oversight moment by moment produces the least data about it over time.

Data Control and Cost Sensitivity Shape Adoption Willingness

Every participant gave a clear position on data storage and cost. What was striking was not the positions but their independence: privacy caution and cost caution did not travel together. IP1 wanted local storage and still accepted a monthly subscription with no proof required first. IP2 cited competitor access as a specific worry, insisted on local-only storage, and remained willing to invest significantly once the tool earned it. E2 wanted a free cloud tool. P1 chose local storage and a small one-time payment with almost no reasoning offered for either. Seven participants, and almost seven

different combinations.

[Ikpe \(2024\)](#) groups resource constraints, technical limits, and data exposure fears as a single adoption barrier cluster for small firms. [Hossain et al. \(2024\)](#) argue that tools earn uptake through demonstrated usefulness. Both hold here, but partially. The data sovereignty codes in Appendix IV were flagged as case-context rather than literature-anchored precisely because the SME adoption literature treats resource constraints, technical limits, and data exposure fears as separate items on a checklist [Ikpe \(2024\)](#), with no attention to whether they move together or apart. In this case they moved apart: participants tolerated cost but not data leaving the country. This sample did not. [Hossain et al. \(2024\)](#)'s proof-of-value logic fits E1 and IP2. It misses E2, who wanted the tool free regardless of performance, and IP1, who was cost-open from the start.

The practical consequence is direct. Assuming privacy-cautious users are also cost-cautious produces one platform tier and gets roughly four of seven participants roughly right. E2's free-cloud position is isolated in this dataset but not uninformative: it marks a floor no subscription model reaches. A local-first, one-time-cost option suits P1, IP3, and IP4. A proof-of-value delay before payment suits E1 and IP2. These are three distinct adoption paths. Storage preference and cost model need to be configurable independently, because the users in this study have already decided them independently.

Succession, Tacit Knowledge, and the Limits of Coordination Models

Grading disagreements were not what participants actually worried about. What surfaced instead was knowledge with no path out of one person's head. P1 wants AI to capture general experience/intuition, hard to put into words," so the business stops depending on a single taster. IP3 said it plainly: I have people with five, six years who still cannot tell me why they gave a grade." Both are owners. Both are describing the same structural problem from opposite ends. Nobody else in the sample named tacit knowledge at all. IP4, despite using the most viscerally sensory language of any participant, did not select it when asked directly.

Design Principle 3 (Table 2.1) holds that a platform must capture unformalised individual expertise over time, not just log outcomes ([Wu et al. 2025](#)). [Ikpe \(2024\)](#) adds that small firms in developing-economy contexts face stacked constraints: limited resources, thin technical capacity, that make this harder in practice. The data confirms the problem. Where it departs from the literature is in who actually cares:

knowledge preservation appeared as an owner-specific anxiety, not a general practitioner concern. [Baumann & Wu \(2023\)](#) address whether managerial hierarchy remains necessary as AI takes over more of the coordinating work; the more AI absorbs that function, the stronger the case for flatter structures becomes. P1 and IP3 frame it as survival planning. Those are not the same thing.

That distinction shapes what the platform should actually do. A coordination-logic system tries to make tacit knowledge transferable across roles. What P1 and IP3 want is a record of one person's reasoning, not a workflow tool, closer to a professional diary that could outlast its author. [Hossain et al. \(2024\)](#) argue that decision support earns adoption by demonstrably helping with the task at hand; for owners, that task is continuity, not throughput. IP4's silence on tacit knowledge sharpens this further: the participant with the richest sensory vocabulary does not believe that vocabulary can be captured, raising a question the platform cannot resolve by design alone.

Where the Literature Runs Out

Sections [5.3](#) to [5.3](#) close six loops between data and literature. But a seventh thing happened, and it belongs in its own space. The study produced findings the literature had not predicted: subjectivity tracking organisational walls rather than individual inconsistency; data sovereignty and cost tolerance moving as independent variables. These were not gaps the literature left open. They are things a deductive design would have closed off before the data had a chance to speak. The inductive approach was not a default. It was the reason these findings were findable at all.

Section [2.10](#) named the fourth gap plainly: not whether AI can grade tea, but what a real trading business actually needs from such a system. The literature had built a coherent case for augmentation over replacement, but had not tested it against a business where grading, pricing, and buyer matching all run through a single person with no formal capture behind them. Section [3.7](#) refused a deductive scheme for the same reason: locking the analysis into pre-specified categories would have contained the findings to what the literature had already prepared for. Both decisions rest on the same diagnosis of what the prior work had not done.

Together, the six themes describe a platform none of the literature had quite proposed: not an automated grader, not a generic recommendation engine, but a calibration record with institutional memory built in. One that anchors comparisons to garden, season, and light conditions; keeps its reasoning visible enough to interrogate; preserves tacit knowledge before it retires with its owner; and treats data

sovereignty and pricing as separate decisions, because practitioners clearly do. Themes 2 and 4 the literature could have anticipated. Themes 1, 5, and 6, in the specific forms they took, it could not. The platform in Section 5.4 follows from those three.

5.4 What the Themes Require the Platform to Do

Figure 5.1 draws the Judgement Boundary the six themes produce together. On the left sits what AI can reasonably systematise: image capture, environmental consistency, recording, comparison, retrieval, buyer matching, price history, and communication preparation. On the right sits what must stay human: smell, taste, sensory interpretation, fault judgement, the final grade, the price decision, and negotiation. The middle is where the two meet, an AI-supported human decision-making space that produces better records and stronger outcomes without shifting authority across the line.

This boundary is not an admission that the technology is not good enough yet. IP4's statement that "half of what I judge never reaches the eye" is an ontological claim, not a complaint about camera resolution. A system that cannot smell or taste cannot grade on those dimensions; any score it generates would be built on incomplete evidence, and a taster leaning on it would not know what they were missing.

What the design space therefore requires: standardised visual capture and recording conditions; every AI output accompanied by a stated reason and a named reference sample; institutional memory stored across tasters, dates, and buyers; and data storage treated as a decision separate from pricing, because participants clearly do. Chapter 6 develops that into a practical proposal.

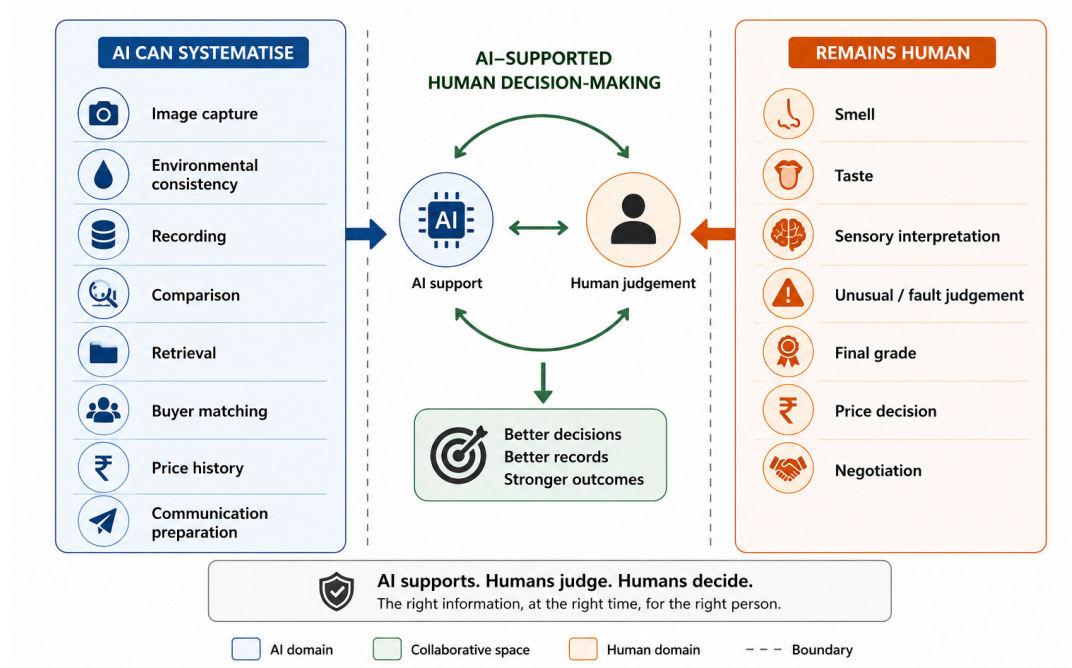


Figure 5.1: Conceptual model of the proposed platform. Six features correspond to the six empirical themes. The human taster remains the decision node at every stage.

5.5 Summary

Six themes emerged from the analysis, each pressing the literature in a different direction. Grading instability tracks organisational boundaries rather than individual inconsistency. Trust in AI is conditional and bounded by IP4's categorical refusal on sensory judgement. AI belongs in the support layer. Human authority over the final grade is unanimous, though the mechanism varies. Data sovereignty and cost tolerance move independently of each other. Tacit knowledge preservation is an owner's concern, not a practitioner one. Chapter 6 draws these findings into recommendations and considers what the study cannot claim.

Chapter 6: Conclusion and Recommendations

6.1 *What Seven Interviews Actually Found*

Answering the Research Question

The study asked what an AI-enabled platform would need to do to support the standardisation of tea tasting and grading in a small Indian brokerage. The answer, compressed, the Judgement Boundary, it would need to systematise everything around the sensory judgement without touching the judgement itself. Grading rests on smell, taste, and contextual reading that no camera or sensor array can hold, and every participant in the study kept the final call with the human taster. A subsequent validation session with the case company confirmed this: final sensory judgement stayed with the taster, and the group identified faster sample preparation and image capture as the most immediate practical gain the system could deliver (full responses in Appendix VII).

What justifies a platform is not grading inconsistency but the operational problems underneath it. Records sit in Excel files, paper diaries, and personal notes with no structured comparison across tasters or over time. Buyer preferences are tracked from memory. Tacit knowledge, the kind two owners described as hard to verbalise, has no path out of one person's head. Data sovereignty and cost tolerance move independently, so no single deployment model fits the whole sample.

The intervention these findings call for is a human-centred assessment and business support system, not an automated grader. Section 6.3 sets out that system in practical terms.

6.2 Contributions of the Research

Table 6.1 sets out what this study adds across four levels. None of the rows is the contribution on its own. What holds them together is a single through-line: participant interviews generated design principles, a validation session with the case company then tested whether those principles held when rendered as a working system proposal, and the result is a recommendation grounded in both. That two-stage pathway is the thing the literature had not attempted.

Table 6.1: Contributions of the study

Level	What this study contributes
Theoretical	Shows that AI adoption in tea assessment is better understood as supporting experiential expertise than replacing it, with trust and accountability remaining human-centred rather than technology-centred
Practical	Translates six empirical themes into a proposed human-AI tea assessment and business support system for Shree Shyaam Tea Co., set out in Section 6.3
Methodological	Demonstrates a two-stage pathway: exploratory interviews generated design principles, and a case-company validation session then tested their practical translation into system requirements within a single study
Contextual	Provides evidence from a small Indian tea trading business, including data locality preferences, tacit knowledge anxieties, buyer matching practices, and informal record keeping, a context the existing literature had not reached

Most dissertations in this space stop at the finding. This one moves from finding to design requirement to proposed solution. The gap between "participants prefer AI as support" and "here is what that support looks like in a working brokerage" is where the contribution sits.

6.3 A Platform That Follows From the Evidence

The six themes point towards one system, not six separate fixes.

6.3.1 The Physical Workflow

The current process (Figure 1.1) asks the taster to handle preparation, photography, and record keeping alongside the sensory work itself. Section 4.2 found all seven participants insisting sensory judgement stays human. Section 4.2 found the real bottleneck in scattered records, not grading speed. Figure 6.1 separates the two. The taster's job narrows to what the findings say it should be: smell, taste, assess, dictate. The enclosed chamber addresses the lighting inconsistency P1 flagged and Section 2.3 treated as structural. The system is designed to fit within the existing approximately 20-minute cycle by running stages in parallel. Whether it achieves that is a question for prototype testing. A validation session with the case company returned positive feedback on this arrangement: faster sample preparation and image capture were identified as the largest anticipated gain.

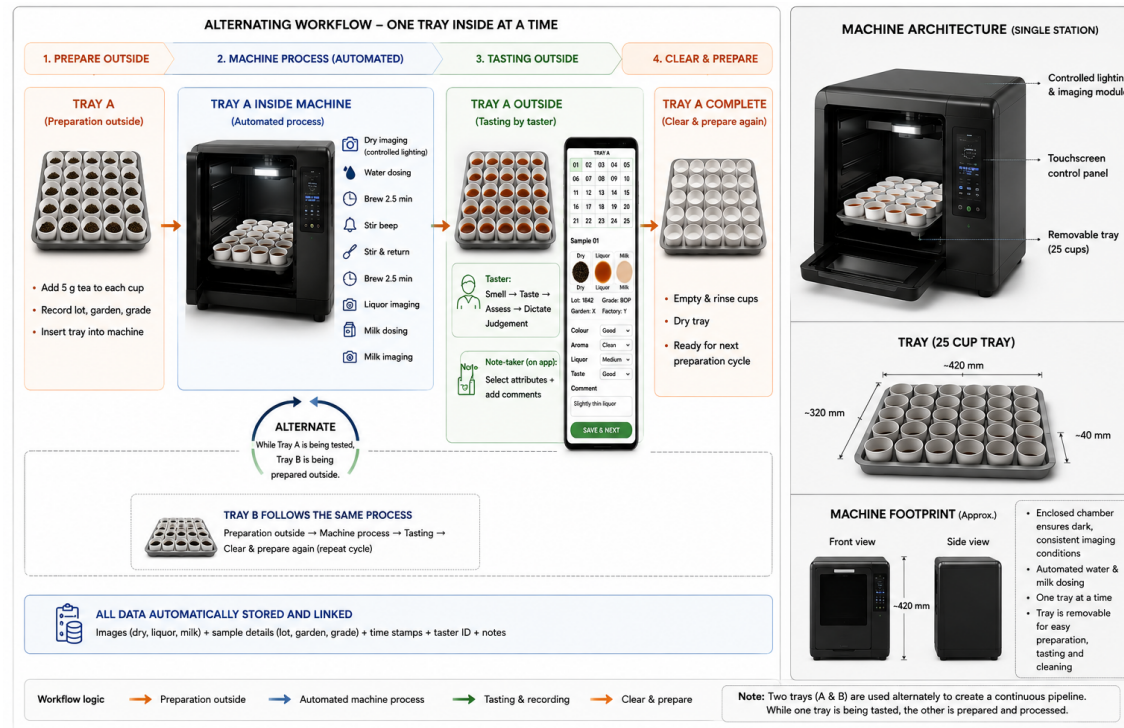


Figure 6.1: Single-station system with alternating 25-sample trays: operational workflow and machine architecture.

6.3.2 The Digital Platform

Figure 6.2 shows the full architecture. Each module traces to a finding. Historical comparison answers Section 4.2's discovery that grade labels shift meaning across firms. Buyer matching addresses the informal, memory-based process described in Section 4.2. The tasting record captures both structured attributes and the taster's own language, preserving the tacit knowledge only P1 and IP3 worried about losing. Storage is local by default, because Section 4.2 found four of seven participants insisting on it. No sensory output is generated, not even as a suggestion, because IP4's objection in Section 4.2 confirmed this was a boundary, not a threshold. Every AI output is a proposal. The trader decides.

The validation session supports AI assistance for colour analysis and tea particle grading, provided the system is trained on business-specific data and framed as image-based decision support rather than autonomous scoring. Note generation was also identified as a preferred AI function.

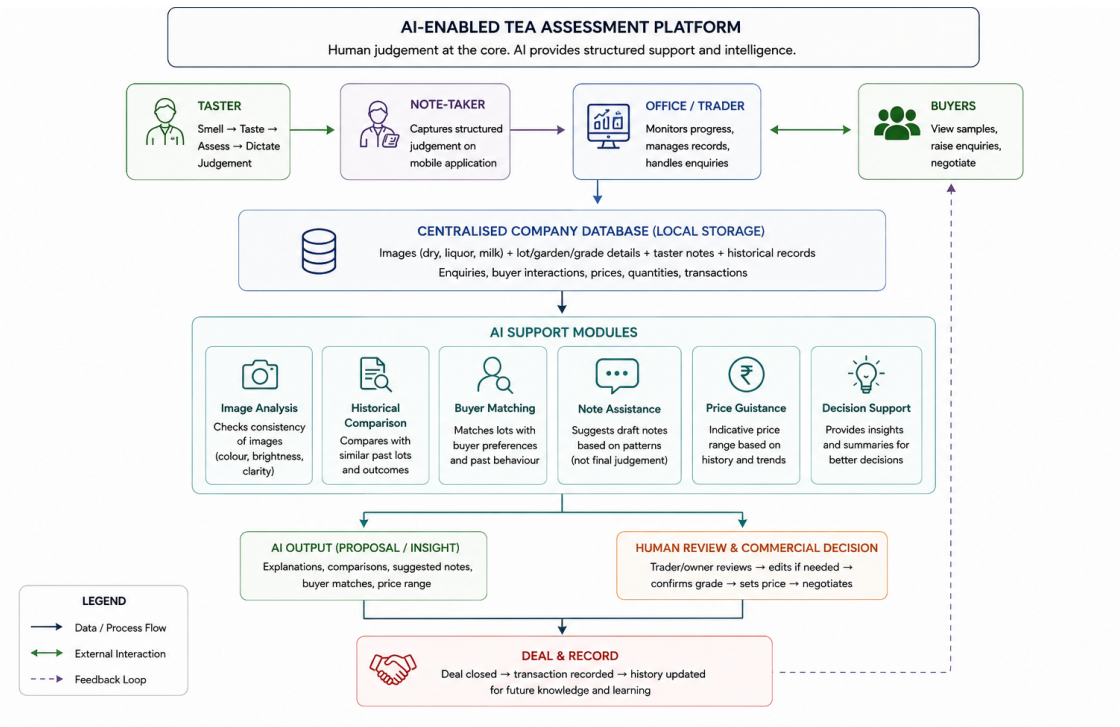


Figure 6.2: Simplified platform architecture for tea assessment and business operations.

6.3.3 Rolling Out in Four Phases

Trust in AI, as Section 4.2 showed, is built, not given. IP3 wanted a month of parallel running. IP1 wanted accuracy proven over time. The recommended rollout moves through four phases: digital records and imaging first (no AI), then historical retrieval, then AI-assisted matching and decision support, then evaluation against trading outcomes.

Phase 1 delivers value on its own terms before any AI component is introduced, fitting the conditional adoption pattern the data described.

For Shree Shyaam specifically, the harder obstacle is unlikely to be the technology but the shift from memory-based practice to structured digital records, a habit change for staff used to informal systems. Starting with digitisation before AI is meant to absorb that resistance gradually rather than force it.

6.4 What This Study Cannot Claim

Six limitations bear on what this study can and cannot claim.

The most obvious is scope. The findings come from one business in Siliguri and seven participants, which gives depth rather than breadth. Transferring these conclusions to the Indian tea trade in general, or to businesses at a different scale, is not what the evidence supports. Recruitment through the proprietor as gatekeeper, and my visible closeness to the case, may also have shaped what participants felt able to disclose.

The sample was small by design. Tea grading is an expert skill, and a random draw was never realistic. Seven participants gave real variation in role and tenure, but whether a larger sample would have altered the themes is genuinely unknown.

Interview evidence was captured through written notes rather than recordings. Note-taking was structured and contemporaneous, but verbatim analysis was not possible and some compression in participants' exact wording was unavoidable.

Participants held different roles and different relationships with grading. That variation was analytically useful and a complication in equal measure. Like-for-like comparison across the sample is harder than the theme-level groupings suggest.

Tea assessment shifts with season, tea type, and market conditions. The study does not account for that variation.

The proposed platform is a design recommendation grounded in participant accounts, not a validated prototype. No claim is made that it would improve grading accuracy, throughput, or commercial outcomes. That work is a different study to this one.

The design-validation session involved three participants from the focal organisation, all of whom had already been interviewed individually. Notes were taken collectively, which means individual agreement or disagreement cannot be separated from the group response. The proprietor-employee relationship within the session may also have shaped how openly participants challenged the proposed design. The session provides case-company validation, not independent confirmation.

The focus group evaluated the proposed design conceptually, not as a working prototype. The reported assessment of "somewhat useful" and the conditional willingness to adopt the system if it worked as shown are not evidence of actual usability, grading accuracy, or performance improvement. That assessment requires a different study.

6.5 *What Needs to Be Built and Tested Next*

The proposed platform needs building and testing before its claims can be assessed. AI outputs should be compared against expert grading on real lots, throughput timed under commercial conditions, and usability tested with practising tasters. Multi-firm work across at least one trading season would establish whether the findings transfer beyond one brokerage.

The validation session also noted that additional requirements would likely surface through trial and error on features currently handled manually. This reinforces the case for iterative prototyping: staged testing should begin with the physical workflow and image capture system before any AI-assisted analysis is layered in.

6.6 *Reflection*

I started this project assuming AI could fix grading. Seven interviews in, that assumption looked naive. The problem was never grading, it was scattered records, undocumented expertise, and a business that runs through one person's memory. Listening to participants did not change the answer so much as rewrite the question I had been asking.

That rewrite was not comfortable. I kept reaching for automation-first framings out of habit and had to consciously resist writing according to what I had planned rather than the one the data supported. The validation session was the real test: watching the case company respond to a design built from their own words, not my assumptions, taught me more about doing qualitative research honestly than any single interview had, and left the harder question unanswered: how much of a taster's judgement outlives the taster.

Bibliography

- Abayomi, O. B. & Noordeen, A. R. (2025), Trusting the black box - dark side of ai risks and ethical concerns, *in* 'Proceedings of the 2025 Pre-ICIS SIGDSA Symposium', p. 73.
URL: aisel.aisnet.org/sigdsa2025/73
- Baumann, O. & Wu, B. (2023), 'Managerial hierarchy in ai-driven organizations', *Journal of Organization Design* **12**.
- Bhattacharyya, N. et al. (2008), 'Electronic nose for black tea classification and correlation of measurements with "tea taster" marks', *IEEE Transactions on Instrumentation and Measurement* **57**(7), 1313–1321.
- Braun, V. & Clarke, V. (2021), 'One size fits all? what counts as quality practice in (reflexive) thematic analysis?', *Qualitative Research in Psychology* **18**(3), 328–352.
- Celestin, M. & Sujatha, S. (2024), 'Ai's impact on corporate strategies and efficiency: Risks, opportunities, and ethical considerations', *Indo American Journal of Multidisciplinary Research and Review* **8**(2), 111–119.
- Chang, T.-S. & Bau, D.-Y. (2025), 'explainable artificial intelligence (xai) in business management research: a success/failure system perspective', *Journal of Electronic Business & Digital Economics* **4**(1), 36–53.
- Corvelo Benz, N. L. & Gomez Rodriguez, M. (2023), Human-aligned calibration for ai-assisted decision making, *in* 'Advances in Neural Information Processing Systems 36'.
- Davis, F. D. (1989), 'Perceived usefulness, perceived ease of use, and user acceptance of information technology', *MIS Quarterly* **13**(3), 319–340.
- Gamoura, S. C. (2023), *Explainable AI (XAI) for AI-Acceptability: The Coming Age of Digital Management 5.0*.

- Gefen, D., Karahanna, E. & Straub, D. W. (2003), 'Trust and tam in online shopping: An integrated model', *MIS Quarterly* **27**(1), 51–90.
URL: [jstor.org/stable/30036519](https://www.jstor.org/stable/30036519)
- Gharibzahedi, S. M. T. et al. (2022), 'Electronic sensor technologies in monitoring quality of tea: A review', *Biosensors* **12**(5), 356.
- Gill, G. S., Kumar, A. & Agarwal, R. (2013), 'Nondestructive grading of black tea based on physical parameters by texture analysis', *Biosystems Engineering* **116**, 198–204.
- Guo, J. et al. (2025), 'Tea grading, blending, and matching based on computer vision and deep learning', *Journal of the Science of Food and Agriculture* **105**, 3239–3251.
- Hossain, M. A. et al. (2024), 'Applying the technology acceptance model (tam) in information technology system to evaluate the adoption of decision support system', *Journal of Computer and Communications* **12**, 242–256.
- Ikpe, E. O. (2024), 'Adoption and implementation of artificial intelligence in small businesses in selected developing countries', *Journal of Health, Applied Sciences and Management* **8**(1), 26–34.
- Laddi, A., Prakash, N. R. & Kumar, A. (2014), 'Quality evaluation of black ctc teas based upon seasonal variations', *International Journal of Food Science and Technology* **49**, 493–500.
- Laddi, A. et al. (2012), 'Significant physical attributes affecting quality of indian black (ctc) tea', *Journal of Food Engineering* **113**, 69–78.
- Lazaros, K., Vrahatis, A. G. & Kotsiantis, S. (2026), 'Human-in-the-loop artificial intelligence: A systematic review of concepts, methods, and applications', *Entropy* **28**(4), 377.
- Ma, S. et al. (2023), Beyond recommender: Effects of different ai roles in ai-assisted decision making, in 'Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems', pp. 1–19.
- Ouvinha de Oliveira, R. et al. (2021), Ai in food and beverage industry, in 'Artificial Intelligence for Digitising Industry'.
- Peng, Y.-H. et al. (2025), Developing tea quality grading system with ai and gtn-9 measurements, in 'International Conference on Applied System Innovation (ICASI 2025)'.

- Pu, Y. et al. (2025), 'The analysis of strategic management decisions and corporate competitiveness based on artificial intelligence', *Scientific Reports* .
- Rahman, M. T. et al. (2021), 'Characterization of tea (*camellia sinensis*) granules for quality grading using computer vision system', *Journal of Agriculture and Food Research* **6**, 100210.
- Ren, G. et al. (2023), 'Rapid characterization of black tea taste quality using miniature nir spectroscopy and electronic tongue sensors', *Biosensors* **13**, 92.
- Retzlaff, C. O. et al. (2024), 'Human-in-the-loop reinforcement learning: A survey and position on requirements, challenges, and opportunities', *Journal of Artificial Intelligence Research* **79**, 359–415.
- Rudko, I., Bonab, A. B. & Bellini, F. (2021), 'Organizational structure and artificial intelligence. modeling the intraorganizational response to the ai contingency', *Journal of Theoretical and Applied Electronic Commerce Research* **16**, 2341–2364.
- Sajeewani, T., Pulasinghe, K. & Marikar, F. (2023), 'Artificial neural network based intelligent tea taster-review', *Automation of technological and business – processes* **15**(1).
- Saunders, M. N. K., Lewis, P. & Thornhill, A. (2023), *Research Methods for Business Students*, 9 edn, Pearson.
- Schröder, A. et al. (2022), Human-ai collaboration: Coordinating automation and augmentation tasks in a digital service company, in 'Proceedings of the 55th Hawaii International Conference on System Sciences', pp. 206–215.
- Sohn, K. & Kwon, O. (2020), 'Technology acceptance theories and factors influencing artificial intelligence-based intelligent products', *Telematics and Informatics* **47**, 101324.
- Tchemeube, R. B., Ens, J., Plut, C., Pasquier, P., Safi, M., Grabit, Y. & Rolland, J.-B. (2025), 'Evaluating human-ai interaction via usability, user experience and acceptance measures for mmm-c: A creative ai system for music composition', *arXiv preprint arXiv:2504.14071* .
- Vudugula, S. et al. (2023), 'Integrating artificial intelligence in strategic business decision-making: A systematic review of predictive models', *International Journal of Scientific Interdisciplinary Research* **4**(01), 01–26.
- Wu, T. et al. (2025), 'Applications of deep learning in tea quality monitoring: a review', *Artificial Intelligence Review* **58**, 342.

Yang, Z. et al. (2023), 'The application status and trends of machine vision in tea production', *Applied Sciences* **13**, 10744.

Zhi, S. et al. (2025), 'Recent advances and applications of imaging and spectroscopy technologies for tea quality assessment: A review', *Agronomy* **15**, 1507.

Appendix I: ERGO Ethics Documentation

I.1 ERGO Ethics Approval

ETHICS APPLICATION FORM

Faculty of Social Sciences

Please note:

- **You must not begin data collection for your study until ethical approval has been obtained.**
- **It is your responsibility to follow the University of Southampton's Ethics Policy (<https://www.southampton.ac.uk/about/governance/policies/ethics.page>) and any relevant academic or professional guidelines in the conduct of your study. This includes providing appropriate information sheets and consent forms, and ensuring confidentiality in the storage and use of data.**
- **You are advised to read the Advice on Applying guidance document, downloadable from the ERGO II website, before you submit your application.**

Important notice on Risk Assessment:

Health and Safety-type risk assessment is no longer part of the ethics review process. Questions pertaining to ethical and reputational risks have been moved from the old 'Risk Assessment Form for Assessing Ethical and Research Risks' to this form. Please do NOT upload a separate Risk Assessment Form to your ethics application.

However, it is your responsibility to undertake a Risk Assessment for your research study. Depending on whether your study is office based, involves off-site data collection and/or international travel, there are different risk assessment forms you can use. Please use this link to access the forms:

<https://groupsite.soton.ac.uk/Administration/FSHS-Health-and-Safety/Documents/Forms/AllItems.aspx?RootFolder=%2FAdministration%2FFSHS%2DHealth%2Dand%2DSafety%2FDocuments%2FRisk%20assessments%20and%20risk%20register%2FERGO%20interim%20documents&FolderCTID=0x012000BE79A4A3B3DC1143ABB38DFA6B580A8C&View={A5E79215-986A-4471-8CF9-B11F85214687}>

If you need guidance or are unsure about which form to use, please contact your Discipline Health and Safety Rep in the first instance, and the Faculty Health and Safety Officer, Aloma Hack (A.J.Hack@soton.ac.uk), if you have further questions. Supervisors and Line Managers are responsible for ensuring risk assessments are completed for all research studies.

1. **Name(s):** Akancha Sharma
2. **Current Position:** Postgraduate Taught Student
3. **Contact Details:**
Division/School: Southampton Business School
Email as21u25@soton.ac.uk
Phone +44 7424443929

4. Is your study being conducted as part of an education qualification?

Yes ☒ No ☐

5. If Yes, please give the name of your supervisor

Andrew Stainton

6. Title of your project:

What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

7. Briefly describe the rationale, study aims and the relevant research questions of your study

This study aims to explore what design features an AI-enabled platform could include to support the standardisation of tea tasting and grading in tea trading businesses. The research focuses on understanding how tea quality is currently assessed, identifying sources of inconsistency in grading practices, exploring perceptions of AI-supported decision making, and developing recommendations for platform design.

Research Question: What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

Objectives:

1. To understand current tea tasting and grading practices.
2. To identify factors that contribute to inconsistent quality assessments.
3. To explore stakeholder perceptions of AI-assisted decision support.
4. To develop practical design recommendations for an AI-enabled platform.

8. Describe the design of your study

I am conducting a qualitative study using semi-structured interviews. The research participants are tea traders, tea buyers, tea brokers, and individuals with direct experience of tea quality evaluation and trading. This study will use purposive sampling and the expected sample size is approximately 5 to 10 participants.

The interviews will be conducted individually via WhatsApp voice or video call, depending on the participant's preference. Participants are based in India and routinely use WhatsApp for professional communication, making it the most practical and accessible platform. Interviews may be conducted in English or Hinglish (a combination of Hindi and English), depending on the participant's preference. No audio or video recordings will be made. Instead, the researcher will document participants' responses through detailed written notes taken during the interview. These written notes will constitute the research data and will subsequently be analysed using thematic analysis.

Participants will be recruited through professional contacts within the tea trading industry, including introductions facilitated by the owner of Shree Shyaam Tea Co. and other industry contacts.

Before conducting the interviews, participants will be provided with a Participant Information Sheet and Consent Form and will have the opportunity to ask questions before providing informed consent.

9. Who are the research participants?

Tea traders, tea buyers, tea brokers, and other tea industry professionals who are professional contacts of the owner will be invited to participate. All participants will have a minimum of five years of experience in tea tasting and grading within the industry. Participants will be adults aged 18 years or over and will be based in India.

10. If you are going to analyse secondary data, from where are you obtaining it?

No.

11. If you are collecting primary data, how will you identify and approach the participants to recruit them to your study?

Participants will be identified through professional contacts within the tea trading industry, personal contacts, direct approaches to industry professionals, and introductions facilitated through the owner of Shree Shyaam Tea Co. acting as a gatekeeper where appropriate.

12. Will you be collecting Special Category data as defined by UK data protection legislation? Will you be collecting Criminal Offence data? If so, please give details.

No.

13. Where will your data collection take place?

Data collection will take place online via WhatsApp voice or video calls. Participants are based in India and WhatsApp is the communication platform routinely used within their professional networks, making it the most practical platform for conducting interviews.

14. Will participants be taking part in your study without their knowledge and consent at the time (e.g. covert observation of people)? If yes, please explain why this is necessary.

No

15. If you answered 'no' to question 14, how will you obtain the consent of participants?

Participants will provide informed consent through the University's information sheet and consent form.

16. Is there any reason to believe participants may not be able to give full informed consent? If yes, what steps do you propose to take to safeguard their interests?

No.

17. If participants are under the responsibility or care of others (such as parents/carers, teachers or medical staff), what permission do you have to approach the participants to take part in the study?

Not applicable.

18. Describe what participation in your study will involve for study participants.

Participants will take part in a semi-structured interview lasting approximately 30–45 minutes. They will be asked about current tea tasting and grading practices, challenges in quality assessment, sources of inconsistency, and their views on the potential use of AI-enabled decision support systems within tea trading.

19. How will you make it clear to participants that they may withdraw consent to participate at any point during the research without penalty?

Participants will be informed of their right to withdraw at any time before, during, or up to 14 days after their participation without giving a reason through the Participant Information Sheet and Consent Form. If a participant withdraws within this period, their interview notes and any identifiable data relating to them will be deleted and will not be included in the study. If they withdraw later than 14 days after their participation, the information already collected may continue to be used for the purposes of achieving the objectives of the study.

20. Detail any possible distress, discomfort, inconvenience, harm or other adverse effects the participants may experience, including after the study, and how you will deal with this.

No significant risks are anticipated. Participants may experience minor inconvenience from the time required to complete the interview. Participants may decline to answer any question and may stop the interview at any time without penalty.

21. Specify any possible distress or harm to YOU arising from your proposed research, and the precautions you will take to minimise these.

No significant risks are anticipated. Appropriate precautions will be taken when conducting interviews online or in person, and all University guidance relating to research activities will be followed.

22. Does your planned research pose any additional risks as a result of the sensitivity of the research and/or the nature of the population(s) or location(s) being studied?

Give considerations to aspects such as impact on the reputation of your discipline or institution; impact on relations between researchers and participants, or between population sub-groups; social, religious, ethnic, political or other sensitivities; potential misuse of findings for illegal, discriminatory or harmful purposes; potential harm to the environment; impacts on culture or cultural heritage.

No significant additional ethical or reputational risks are anticipated. The study focuses on professional experiences relating to tea trading and grading rather than sensitive personal matters.

23. How will you maintain participant anonymity and confidentiality in collecting, analysing and writing up your data?

Participants will be assigned pseudonyms or participant codes. Any identifying information will be removed from interview notes and research reports. Only the researcher and supervisor will have access to identifiable data. Findings will be reported in a way that prevents individual participants from being identified.

24. How will you store your data securely during and after the study?

All information collected will be stored securely on University OneDrive and other approved University of Southampton storage systems., under My Documents on the University of Southampton file storage space, or on Microsoft OneDrive/Teams/SharePoint/Research Data accessed using their University email address. No audio or video recordings will be stored because interviews will not be recorded.

25. Describe any plans you have for feeding back the findings of the study to participants.

Participants may request a summary of the findings after completion of the dissertation.

26. What are the main ethical issues raised by your research and how do you intend to manage these?

I have received training on ethical issues in the lectures of MANG6550 Dissertation module, and have access to ethics guidance on Blackboard. Any ethical issues/concerns related to my study will be included in my research proposal, final dissertation, and discussed with my supervisor. Any unaddressed ethical concerns or disputes will be raised with the module leader.

27. Please outline any other information you feel may be relevant to this submission.

None.

I.2 Participant Information Sheet

Participant Information Sheet

Study Title: What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

Researcher: Akancha Sharma

Ethics/ERGO number: 115009

You are being invited to take part in the above study. Please read the information below carefully and ask questions if anything is not clear or you would like more information before you decide to take part in this research. If you are happy to participate you will be asked to give your consent.

What is the research about?

My name is [Akancha Sharma](#), and I am studying for an MSc in [Management and Artificial Intelligence](#) at the University of Southampton in the United Kingdom.

I invite you to participate in a study [exploring what design features an AI-enabled platform could include to support the standardisation of tea tasting and grading in tea trading businesses](#).

The study aims to understand how tea quality is currently assessed, identify sources of inconsistency in grading practices, explore views on AI-assisted decision support, and develop recommendations for a platform that could support more consistent tea evaluation.

Why have I been asked to participate?

You have been invited to participate because you are a tea trader, tea buyer, tea broker, or tea industry professional with at least five years of experience in tea tasting and grading. I aim to recruit approximately [5 to 10 participants](#) for this study.

What will happen to me if I take part?

If you participate, you will take part in a semi-structured interview lasting approximately 30-45 minutes. You will be asked questions about tea tasting and grading practices, challenges in quality assessment, sources of inconsistency, and your views on the potential use of AI-enabled decision support systems.

The interview will be conducted via WhatsApp voice or video call. Depending on your preference, the discussion may take place in English or Hinglish (a combination of Hindi and English). No audio or video recording will be made. Instead, the researcher will take detailed written notes throughout the interview, which will form the research data for analysis.

Are there any benefits in my taking part?

There are no direct benefits to participating in this study. However, your responses will contribute to knowledge in this research area.

Are there any risks involved?

No risks are anticipated.

What data will be collected?

Data will be collected about your experiences, opinions, and professional practices relating to tea tasting, grading, quality assessment, and the potential use of AI-supported decision-making tools within tea trading businesses.

How and where data will be stored?

All information collected will be stored securely on University of Southampton approved storage systems, including OneDrive and other secure University platforms. As no audio or video recordings will be made, only the written interview notes will be retained and accessible only to the researcher and supervisor.

All data collected for the purpose of this study will be securely destroyed after the conferment of my degree.

Will my participation be confidential?

Your participation and the information I collect about you during the course of the research will be kept confidential. Only the researcher and their supervisor will have access to the data. In addition, responsible members of the University of Southampton may be given access for monitoring purposes to ensure compliance with applicable regulations. They have a duty to keep your information confidential.

To protect your identity, I will assign individual participants ID numbers. Only the researcher and their supervisor will have access to the codes.

Do I have to take part?

No, it is entirely up to you to decide whether to take part. If you decide that you want to participate, you will need to *sign a consent form* to show that you have agreed to take part.

What happens if I change my mind?

You have the right to change your mind and withdraw at any time during the study without giving a reason.

You may withdraw from the study at any time before, during, or up to 14 days after your participation without giving a reason. If you withdraw within this period, your interview notes and any identifiable data relating to you will be deleted and will not be included in the study.

If you withdraw later than 14 days after your participation from the study, we will keep the information about you that we have already obtained for the purposes of achieving the objectives of the study only.

What will happen to the results of the research?

The information you provide me with will be analysed and written up as part of my dissertation. Your personal details will remain confidential. Research findings in my dissertation or any report arising from it will not include information that can directly identify you without your specific consent.

Where can I get more information?

Name: Akancha Sharma

Email: as21u25@soton.ac.uk

Supervisor: **Andrew Stainton**

What happens if there is a problem?

If you have a concern or a complaint about any aspect of this study, please contact the Head of Research Integrity and Governance, University of Southampton, on the following contact details: email: rgoinfo@soton.ac.uk; tel: +44 23 8059 5058.

Thank you.

Thank you for taking the time to read this information sheet and considering taking part in this research.

Data Protection Privacy Notice

How will we use information about you?

For the purposes of data protection law, the University of Southampton is the 'Data Controller' for this study, we will need to use information from you for this research project.

This information will include your name and email address. People will use this information to do the research or to check your records to make sure that the research is being done properly. People who do not need to know who you are will not be able to see your name or email address. Your data will have a code number instead. We will keep all information about you safe and secure.

Once we have finished the study, we will keep some of the data so we can check the results. We will write our reports in a way that no-one can work out that you took part in the study.

What are your choices about how your information is used?

- You can stop being part of the study at any time, without giving a reason, but we will keep all anonymised information about you that we already have.
- We need to manage your records in specific ways for the research to be reliable. This means that we won't be able to let you see or change the data we hold about you.

Where can you find out more about how your information is used?

You can find out more about how we use your information:

- by sending an email to University's Data Protection Officer (data.protection@soton.ac.uk).
- by asking one of the research team or from our [general privacy policy](#).
- by sending an email to as21u25@soton.ac.uk, or by ringing us on +44 7424443929.

Thank you for taking the time to read the information sheet and considering taking part in the research.

I.3 Consent Form

CONSENT FORM

Study Title: What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

Ethics/ERGO Number: 115009

Version and date: V2 25.06.2026

Thank you for your interest in this study. It is very important to us to conduct our studies in line with ethics principles, and this Consent Form asks you to confirm if you agree to take part in the above study. Please carefully consider the statements below, and add your initials and signature only if you agree to participate in this research and understand what this will mean for you.

Please add your initials to the boxes below if you agree with the statements:

Mandatory Consent Statements	Participant Initials
I confirm that I read or listened to the Participant Information Sheet version V2, dated 25.06.2026, explaining the study above and I understand what is expected of me.	
I was given the opportunity to consider the information, ask questions about the study, and all my questions have been answered to my satisfaction.	
I agree to take part in this study and understand that data collected during this research project will be used for the purpose of this study.	
I understand that my participation is voluntary and that I am free to withdraw from this study at any time without giving a reason.	

Additional Statements - please add your initials in the boxes below you to agree to:

Additional Consent Statements	Participant Initials
I understand that I may withdraw from the study at any time before, during, or up to 14 days after my participation without giving a reason. If I withdraw within this period, my interview notes and any identifiable data relating to me will be deleted and will not be included in the study.	
I understand that if I withdraw later than 14 days after my participation, the information already collected about me may continue to be used for the purposes of achieving the objectives of the study.	

Name of participant

Signature

Date

Name of person taking consent

Signature

Date

*Once this Consent Form has been signed by all parties, a copy of the signed and dated form should be provided to the study participant. Original signed copy should be stored in the study site file (If applicable).

I.4 Gatekeeper Letter

ERGO Number: **115009**

Date: 25 June 2026

Mr Madhav Sharma
Proprietor
Shree Shyaam Tea Co.
Siliguri, West Bengal
India
Email: madhav673@gmail.com
Phone: +91 9933487398

Subject: Request to Act as Gatekeeper for Research Study

Dear [Mr Sharma](#),

My name is Akancha Sharma. I am a student at the University of Southampton. As part of my Master's degree, I am conducting research on how tea quality is assessed within tea trading businesses and what design features an AI-enabled platform could include to support the standardisation of tea tasting and grading practices under the supervision of Andrew Stainton.

I would like your permission to share information about this study with tea traders, tea buyers, tea brokers, and other tea industry professionals known to you, so that they can decide whether they would like to participate in an interview for this research.

Participation will be entirely voluntary. Choosing not to participate will have no consequences, and no identifying information about participants will be included in the dissertation or any related research outputs.

Thank you for your time and consideration.

Best regards,

[Akancha Sharma](#)
[MSc Management and Artificial Intelligence](#)
[University of Southampton](#)
Email: as21u25@soton.ac.uk

1.5 Recruitment Message

Study Title: [What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?](#)

ERGO Number: **115009**

As part of my master's dissertation at the University of Southampton, I am conducting research on how tea quality is currently assessed within tea trading businesses and what design features an AI-enabled platform could include to support more consistent and standardised tea tasting and grading practices.

If you are a tea trader, tea buyer, tea broker, or tea industry professional with at least five years of experience in tea tasting and grading, I would greatly appreciate the opportunity to interview you to gain a deeper understanding of your views on this topic. The interview will take approximately 30 to 45 minutes. The interview will be conducted via WhatsApp voice or video call. No audio or video recording will be made, and responses will instead be documented through detailed written notes. Interviews may be conducted in English or Hinglish, depending on your preference.

If you are interested in learning more and considering participation, please refer to the attached Participant Information Sheet.

Thank you for your time and support.

Akancha Sharma
as21u25@soton.ac.uk

[\[Participant Information Sheet\]](#)

Appendix II: Interview Questions Asked

Research Questionnaire

What design features of an AI-enabled platform could support the standardisation of tea tasting and grading in tea trading businesses?

Researcher: Akancha Sharma | Supervisor: Andrew Stainton | Ethics/ERGO No: 115009

University email: as21u25@soton.ac.uk

About this study

This research explores how tea quality is currently assessed in tea trading businesses, and what features an AI-enabled platform could include to support more consistent tea tasting and grading. It is part of an MSc dissertation at the University of Southampton, UK.

You are being asked to take part because you are a tea trader, tea buyer, tea broker, or tea industry professional with at least five years of experience in tea tasting and grading.

Taking part is entirely voluntary. There are no direct benefits or risks to you from taking part. You do not have to answer any question you do not want to, and you may stop at any point without giving a reason.

Your answers will be kept confidential. Only the researcher and supervisor will see your name; your responses will be given a code number and your name will not appear in the dissertation or any related report. Your data will be stored securely on University of Southampton approved storage and destroyed after the degree is awarded.

You may withdraw at any time before, during, or up to 14 days after returning this form, without giving a reason. If you withdraw within this period, your answers will be deleted and not used. If you withdraw later than 14 days, the information already collected may still be used for the study.

If you have a concern about this study, you may contact the Head of Research Integrity and Governance, University of Southampton: rgoinfo@soton.ac.uk / +44 23 8059 5058.

Section A: About You

Q1. What is your current role in the tea industry?

- ☐ Owner / Proprietor
- ☐ Tea Taster / Grader
- ☐ Tea Trader
- ☐ Tea Buyer
- ☐ Tea Broker
- ☐ Other, please specify: _____

Q2. How many years have you been involved in tea tasting, grading, buying, or trading?

- ☐ 5–10 years
- ☐ 11–15 years
- ☐ 16–20 years
- ☐ More than 20 years

Q3. What are your main responsibilities relating to tea quality evaluation? *(tick all that apply)*

- ☐ Tasting and grading tea
- ☐ Making buying decisions
- ☐ Price negotiation
- ☐ Managing buyer relationships
- ☐ Record keeping
- ☐ Other, please specify: _____

Section B: Current Tea Tasting and Grading Practices

Background: Studies on tea and food quality assessment often note that sensory judgements — colour, aroma, taste, and appearance — can vary between evaluators, and sometimes between regions or gardens.

The following questions explore how quality is defined and applied in your own work.

Q4. How is tea quality currently assessed in your work? (tick all that apply)

- ☐ Colour
- ☐ Aroma
- ☐ Appearance
- ☐ Liquor
- ☐ Leaf size
- ☐ Strength
- ☐ Taste
- ☐ Overall value
- ☐ Other, please specify: _____

Q5. How are grading decisions normally made? (tick all that apply)

- ☐ Personal judgement
- ☐ Previous experience
- ☐ Taste memory
- ☐ Organisational standards
- ☐ Buyer requirements
- ☐ Other, please specify: _____

Q6. How are tasting records or grading information currently documented and stored? (tick all that apply)

- ☐ Diaries
- ☐ Excel
- ☐ Personal notes
- ☐ Paper records
- ☐ Digital system
- ☐ Not recorded
- ☐ Other, please specify: _____

Q7. When you say a tea is "BOP" or "PF" quality, how much of that is a fixed standard versus specific to a garden, season, or region you know well?

- ☐ It's a fixed industry standard
- ☐ Mostly fixed, with minor regional variation
- ☐ It varies significantly by garden or region
- ☐ I mainly rely on my own judgement, regardless of the label

Q8. Have you seen the same grade label mean a different quality level from a different garden or region? (tick all that apply)

- ☐ I haven't experienced this
- ☐ Particle size / mechanical grade definition differs
- ☐ Overall strength differs
- ☐ Colour depth differs
- ☐ Aroma expectations differ
- ☐ Other, please specify: _____

Q9. How closely does your quality score translate directly into the price you'd quote?

- ☐ Very directly
- ☐ Roughly, as a guide
- ☐ Mostly gut feel

Section C: Environmental and Operational Challenges

Background: Research on quality assessment technology is often conducted under fixed, controlled lighting. In practice, trading businesses operate under changing natural light, at different times of day, and often under time pressure.

The following questions explore the practical, everyday conditions you work in.

Q10. What factors do you think create inconsistency in tea quality evaluation? (tick all that apply)

- ☐ Human judgement
- ☐ Experience level
- ☐ Environmental conditions / lighting / weather
- ☐ Different tasters involved
- ☐ Buyer expectations
- ☐ Other, please specify: _____

Q11. Have you experienced situations where different people assigned different grades to the same tea? If so, what do you think was the main reason? (tick all that apply)

- ☐ I haven't experienced this
- ☐ Different taster's palate / experience
- ☐ Different lighting or time of day
- ☐ Different batch or sample age
- ☐ No particular reason known
- ☐ Other, please specify: _____

Q12. Do you have a preferred time of day, room setup, or method you use to reduce the effect of changing light on your assessment? (tick all that apply)

- ☐ Morning natural light
- ☐ A specific spot, e.g. near a window
- ☐ Consistent electric lighting
- ☐ No particular method
- ☐ Other, please specify: _____

Q13. What conditions make visual assessment hardest for you? (tick all that apply)

- ☐ Dust
- ☐ Humidity
- ☐ Poor lighting
- ☐ Moving / handled stock
- ☐ Other, please specify: _____

Q14. On a busy trading day, roughly how long can you spend assessing one sample before it slows things down?

- ☐ Under 1 minute
- ☐ 1–5 minutes
- ☐ 5+ minutes

Section D: Buyer Relationships and Sensory Judgement

Background: Quality disagreements between traders and buyers are frequently reported as a reason deals do not proceed. Sensory evaluation also often involves weighing several senses — sight, smell, and taste — against each other.

The following questions explore both of these areas.

Q15. Have differences in quality assessment ever affected business decisions, buyer relationships, negotiations, or completed deals?

- ☐ Often
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

Q16. Is there a single fault that can disqualify a lot on its own, regardless of everything else looking fine?
(tick all that apply)

- ☐ Dull or flat liquor colour
- ☐ Off / musty aroma
- ☐ Weak strength or thin liquor
- ☐ Excess stalk or fibre content
- ☐ Inconsistent particle size
- ☐ No single factor — depends on the overall picture
- ☐ Other, please specify: _____

Q17. If colour looks excellent but aroma is only average, how do you usually resolve that?

- ☐ Colour usually wins
- ☐ Aroma usually wins
- ☐ Depends on the buyer's priority
- ☐ I weigh them equally
- ☐ Other, please specify: _____

Section E: Artificial Intelligence in Tea Tasting and Grading

Background: Recent studies have explored the use of Artificial Intelligence in food and agricultural quality assessment. These systems can analyse characteristics such as colour, texture and appearance, while experienced professionals continue to make the final judgement. Researchers are now investigating whether similar approaches could support other industries where quality assessment relies heavily on expert experience.

The following questions explore your views on these possibilities.

Q18. Before today, had you heard about AI being used to support quality assessment in food or agriculture?

- ☐ Yes
- ☐ No

Q19. Knowing that AI can assist experts rather than replace them, what are your thoughts on using AI in tea tasting and grading?

- ☐ Very positive
- ☐ Positive
- ☐ Neutral
- ☐ Sceptical
- ☐ Very sceptical

What best describes why you feel this way? (tick all that apply):

- ☐ Could reduce disputes with buyers
- ☐ Could save time

- ☐ Could help preserve tea-tasting knowledge for the future
- ☐ Not confident it can match human judgement for tea
- ☐ Concerned about cost
- ☐ I don't know enough about it yet
- ☐ Other, please specify: _____

Q20. Which parts of the tea evaluation process do you think could benefit from AI support? *(tick all that apply — if more than one, please circle your single top choice)*

- ☐ Image analysis of tea samples
- ☐ Comparing with historical samples
- ☐ Record keeping
- ☐ Matching tea lots to buyer preferences
- ☐ Price guidance
- ☐ Writing quality descriptions / tasting notes
- ☐ General decision support
- ☐ None
- ☐ Other, please specify: _____

Q21. Are there any parts of tea tasting or grading that should always remain under human judgement? *If yes, which? (tick all that apply)*

- ☐ No, none in particular
- ☐ Final price decision
- ☐ Buyer relationship / negotiation
- ☐ Judging aroma or taste
- ☐ Judging unusual or unfamiliar lots
- ☐ Other, please specify: _____

Q22. Do you have any concerns about using AI within tea trading? *(tick all that apply — if more than one, please circle your single biggest concern)*

- ☐ Accuracy
- ☐ Trust
- ☐ Cost
- ☐ Ease of use
- ☐ Privacy
- ☐ Reliability
- ☐ None
- ☐ Other, please specify: _____

Q23. What would an AI need to show you — not just tell you — for you to trust its grade? *(tick all that apply)*

- ☐ A clear reason or explanation for the grade given
- ☐ A comparison to a past sample I recognise
- ☐ Accuracy proven over time
- ☐ Ability to review and override it myself
- ☐ Other, please specify: _____

Q24. If an AI disagreed with your grading on a lot, what would you want to happen next?

- ☐ I override it silently
- ☐ Flag it for review
- ☐ It learns from my correction
- ☐ Other, please specify: _____

Section F: Platform Design, Buyer Matching, and Knowledge Preservation

Background: Several studies have explored AI as a decision-support tool that assists experts by storing records, comparing previous cases, and organising information, rather than making decisions independently. Trading businesses also often rely on individual memory for buyer preferences and years of comparison points.

The following questions explore whether this kind of support would be useful in your work.

Q25. If an AI-enabled platform were developed for tea trading businesses, which features would you find most useful? (tick all that apply — if more than one, please circle your single top choice)

- ☐ AI-assisted quality grading (image based)
- ☐ Lighting / environment correction for photos
- ☐ Historical comparison of samples
- ☐ Buyer preference matching
- ☐ Price guidance
- ☐ Auto-generated tasting notes
- ☐ Other, please specify: _____

Q26. What information should such a platform store to improve consistency in tea tasting and grading? (tick all that apply)

- ☐ Tasting records
- ☐ Buyer preferences
- ☐ Price history
- ☐ Sample images
- ☐ Grading criteria / standards
- ☐ Other, please specify: _____

Q27. When a buyer says something like "I want the same profile as the July shipment," how is that currently understood and remembered? (tick all that apply)

- ☐ I personally remember the sample
- ☐ Noted in a diary / notebook
- ☐ Sample physically kept for reference
- ☐ Discussed verbally with the team
- ☐ No consistent method
- ☐ Other, please specify: _____

Q28. Do you keep any specific "anchor" samples or remembered lots that you compare new tea against?

- ☐ Yes
- ☐ No

If yes, roughly how far back do these reference samples go?

- ☐ Less than 1 year
- ☐ 1–5 years
- ☐ 5–10 years
- ☐ More than 10 years

Q29. What specific piece of your own knowledge would you most want an AI system to capture, so the business isn't dependent on one person? (tick all that apply)

- ☐ How to judge liquor colour and clarity
- ☐ How to detect off aromas or faults
- ☐ How to match tea to buyer preference
- ☐ How to price a lot appropriately

- ☐ General experience / intuition — hard to put into words
- ☐ Other, please specify: _____

Section G: Technology and Accessibility

Background: Adoption of new technology often depends on whether it is genuinely useful without creating extra burden, particularly for smaller businesses where advanced equipment may not always be practical.

The following questions explore what would be usable and accessible for a business like yours.

Q30. Do you currently use any digital systems or software to support tea tasting, grading, or record keeping? If yes, what do you use? (tick all that apply)

- ☐ No, I don't
- ☐ Excel
- ☐ Paper-based system
- ☐ Mobile notes / photos
- ☐ Other, please specify: _____

Q31. How are buyer preferences currently remembered or managed? (tick all that apply)

- ☐ Personal memory
- ☐ Written notes
- ☐ Verbal communication within the team
- ☐ No formal system
- ☐ Other, please specify: _____

Q32. Would you be comfortable with tasting data going to a cloud platform, or would you prefer it stored locally within the business only?

- ☐ Cloud is fine
- ☐ Local only
- ☐ Depends on what's stored

What best describes your reasoning? (tick all that apply):

- ☐ Concerned about competitors accessing data
- ☐ Prefer data staying within India / the business
- ☐ General privacy preference
- ☐ No particular concern
- ☐ Other, please specify: _____

Q33. Roughly what would feel like a reasonable cost for a tool like this for a business your size?

- ☐ No cost — would need to be free
- ☐ Small one-time cost (e.g. for equipment)
- ☐ Small monthly subscription
- ☐ Willing to invest significantly if it proves valuable
- ☐ Not sure

Section H: Closing

Q34. Is there anything else you think could improve tea tasting, grading, or tea trading that we haven't asked about?

Q35. Is there anything else you'd like to add about the future use of AI in the tea trading industry?

Thank you very much for your time and for sharing your professional experience — it is genuinely valuable to this research.

Appendix III: Company Details: Background and Preliminary Discussion

III.1 Company Background

Shree Shyaam Tea Co. is a tea trading and brokerage business based in Siliguri, India. It sources, evaluates and places tea on behalf of buyers and sellers; it does not grow, process or package tea itself.

Table III.1: Company profile at a glance

Location	Siliguri, West Bengal, India
Trading history	19 years
Sector	Tea trading and brokerage; CTC black tea; not a packaged brand
Revenue model	1% brokerage commission from the seller per completed deal
Customers	Traders, packetiers, exporters (all onward buyers, no end consumers)
Industry network	Tea gardens, factories, traders, packetiers, exporters
Owner's tasting experience	20+ years

Table III.2: The trading process

Step	What happens
Procurement	Tea sourced from gardens, factories, auctions; quantity, price, garden name logged, mostly on paper
Evaluation	Tea tasted and graded to determine quality and market value
Buyer identification	Buyers contacted to establish requirements and price expectations
Lot selection	Suitable lot matched to buyer, based on experience and memory
Negotiation	Terms negotiated between buyer and seller
Deal completion	Transaction documented (fast, largely automated) and closed

Table III.3: CTC grades handled

Grade	Character
BOP	Medium granule; full bodied, balanced cup; everyday staple
BP	Smaller than BOP; stronger, brisker liquor; used in blends
PF	Smaller again; brews quickly; common tea bag grade
PD	Fine, powder like; strong, dark, economical brew; blends and milk tea
Dust / SFD	Finest grade; brews almost instantly; very strong liquor; kadak style

Records are kept in Excel files, diaries and personal notes. They are useful on their own terms but not built for systematic comparison across tasters or over time, the practical starting point for the research question in Chapter 1.

III.2 Preliminary Discussion with the Proprietor

In-person conversation on 6th April, 2026:

Before the proposal was finalised, the researcher held an informal, unrecorded conversation with the proprietor about tasting, grading and trading. No consent process applied, since it predates ethical approval and was never intended as data collection. It is summarised here because it shaped the interview guide (Appendix II) and two initial codes in the TA (Appendix IV).

Table III.4: Themes raised in the discussion

Theme	Summary
Subjective grading	Grading is judgement, not measurement; two tasters can grade the same sample differently; a wrong grade has cost the business money
Environmental effects	Sunlight, cloud and rain change perceived liquor colour and leaf brightness; natural light is preferred but still inconsistent
Buyer/trader mismatch	Trader and buyer readings of the same tea can differ, even in good faith; this sometimes ends deals
Memory dependence	Excel sheets and diaries exist, but comparing samples over time relies on personal recall, not a system
Knowledge preservation	Proprietor open to AI learning from his tasting history, "so that the system can continue even after me"
Data privacy	Open to AI for grading, pricing and buying support, only if data stays inside the business

Table III.5: Case-context codes grounded in this discussion

Code	Basis	Status
DATA-LOCALITY	Proprietor's stated condition for trusting AI with business data	Direct statement
TIME-PRESSURE	Low-stock, demand-driven buying implies grading under commercial tempo	Inference, not a direct statement

Disagreement between tasters, raised repeatedly here, is captured under

SUBJ-DRIFT (Appendix [IV](#)), which already has a literature basis in Chapter [2](#) and needs no separate case-context code.

Appendix IV: Mapping of Literature to Questionnaire Design and Initial Codes

Table IV.1: Literature findings, key sources, initial codes and corresponding questionnaire items

Literature finding / principle	Key literature	LR §	Initial code	Question(s)
Grading rests on human sensory judgement that is not repeatable across sittings or across tasters	(Bhattacharyya et al. 2008 , Laddi et al. 2012 , Sajeewani et al. 2023 , Guo et al. 2025)	2.2	SUBJ-BASIS: basis of a grading decision SUBJ-DRIFT: same tea, different grade	Q4, Q5, Q7, Q8, Q10, Q11
Tea's sensory attributes are shaped by region, season and cultivation before assessment begins; lighting and weather then add further visual variance at the point of assessment	(Wu et al. 2025 , Laddi et al. 2014 , Zhi et al. 2025)	2.3	ENV-LIGHT: lighting and weather effects ENV-WORKAROUND: existing self-correction habits	Q10, Q12, Q13
Commercial assessment happens under time pressure that laboratory and bench-based studies do not model	Case-context (Appendix III.2)	—	TIME-PRESSURE: assessment under commercial tempo	Q14
Computer vision reads surface attributes reliably, struggles on adjacent grades, and cannot capture liquor chemistry	(Rahman et al. 2021 , Gill et al. 2013 , Yang et al. 2023 , Zhi et al. 2025)	2.4	CV-VISIBLE: attributes a camera could capture CV-BLINDSPOT: attributes it could not	Q4, Q16, Q25

Continued on next page

Table IV.1 continued from previous page

Literature finding / principle	Key literature	LR §	Initial code	Question(s)
Aroma, taste and colour interact, and no single instrument captures the whole assessment	(Gharibzahedi et al. 2022, Ren et al. 2023)	2.5	SENSE-TRADEOFF: resolving conflicting cues FAULT-VETO: single disqualifying fault	Q16, Q17
Sensor-based systems assume equipment and technical capacity a small brokerage is unlikely to hold	(Ren et al. 2023, Peng et al. 2025, Ikpe 2024)	2.5, 2.8	RESOURCE-LIMIT: existing technical footprint COST-THRESHOLD: affordable investment level	Q30, Q33
AI in judgement-heavy work performs better as comparison and reference than as verdict; passive acceptance erodes the judgement it is meant to support	(Schröder et al. 2022, Ma et al. 2023, Lazaros et al. 2026, Retzlaff et al. 2024, Tchemeube et al. 2025)	2.6	AI-ROLE: support versus replacement HUMAN-RESERVED: what must stay human OVERRIDE: handling disagreement with the system	Q19, Q20, Q21, Q24
Trust depends on visible reasoning rather than accuracy alone, and opaque output cannot be interrogated	(Gamoura 2023, Abayomi & Noordeen 2025, Corvelo Benz & Gomez Rodriguez 2023, Chang & Bau 2025)	2.7	EXPLAIN-NEED: what the system must show TRUST-EVIDENCE: what would earn confidence	Q23, Q24
Adoption follows perceived usefulness and ease of use, with trust folded into both	(Davis 1989, Sohn & Kwon 2020, Gefen et al. 2003, Hossain et al. 2024)	2.7	TAM-USEFUL: perceived usefulness TAM-EASE: perceived ease of use AI-SCEPTICISM: stated reservations	Q18, Q19, Q22, Q33

Continued on next page

Table IV.1 continued from previous page

Literature finding / principle	Key literature	LR §	Initial code	Question(s)
Expertise concentrated in one person is a preservation risk with no structured capture	(Wu et al. 2025, Ikpe 2024, Baumann & Wu 2023)	2.8	TACIT-KNOWLEDGE: knowledge hard to articulate SUCCESSION: dependence on one individual ANCHOR-SAMPLE: memory used as a reference system	Q27, Q28, Q29
Structured historical data used systematically can outperform decisions made from memory alone	(Vudugula et al. 2023, Pu et al. 2025, Celestin & Sujatha 2024, Hossain et al. 2024)	2.9	RECORD-PRACTICE: current documentation HIST-COMPARISON: comparison against past lots BUYER-MATCH: matching lots to buyer preference PRICE-JUDGEMENT: quality translated into price	Q6, Q9, Q15, Q25, Q26, Q31
Data governance and locality preferences shape willingness to adopt among smaller firms; not addressed directly in the literature reviewed	Case-context (proprietor's stated data-locality preference; not Chapter 2)	—	DATA-LOCALITY: cloud versus local storage PRIVACY-CONCERN: stated reasoning	Q22, Q32
No prior anchor in Chapter 2; reserved for inductive coding	Not applicable	—	Emergent codes only	Q34, Q35

Note. Questions Q1 to Q3 collect participant-profile information and derive from the sampling strategy set out in Section 3.5 rather than from Chapter 2. Two rows above are marked case-context rather than literature-anchored: these questions came from preliminary discussion with the proprietor of Shree Shyaam Tea Co. rather than from Chapter 2, consistent with the inductive approach described in Section 3.2. Codes listed here were the starting frame for analysis and were revised as coding proceeded, as described in Section 3.7.

Appendix V: Participant profiles and key response summary

Participant codes are used throughout the dissertation to protect anonymity.

P = Proprietor; E = Employee; IP = Industry Peer;

V.1 Participant Overview

Table V.1: Overview of participant roles, experience, and notable responses

Code	Group	Role	Experience	Relevant background
P1	Proprietor	Owner / Proprietor	16–20 yrs	Handles tasting, grading, buying and price negotiation himself, and manages buyer relationships directly. Treats grade labels (BOP, PF) as fixed industry benchmarks and prices quite directly off the quality score. Cautiously open to AI, sees value in record-keeping, price guidance, and drafting tasting notes, but would quietly override an AI grade he disagreed with, and wants the final price call and buyer relationship kept human.

continued on next page

Table V.1 – continued from previous page

Code	Group	Role	Experience	Relevant background
E1	Employee	Tea Trader	11–15 yrs	Works mainly on price negotiation and buyer relationships rather than tasting or buying directly. Fairly neutral about AI and not convinced it can match a human palate; his main worry is ease of use. Can imagine it helping with record-keeping and buyer-preference matching, but wants aroma and taste judgement to stay human.
E2	Employee	Tea Taster / Grader	11–15 yrs	Tastes and grades day-to-day, working from personal judgement and taste memory rather than a fixed process. Noted unprompted that "blending is a very important part" of the job, the only participant who wanted to discuss the questionnaire further with the researcher.
IP1	Industry Peer (young professional)	Tea Buyer	5–10 yrs	Responsible for buying decisions and price negotiation. Already aware of AI use in food/agriculture quality assessment; broadly neutral here but likes the idea of it helping preserve tasting knowledge and match lots to buyer preference. Wants buyer relationships and negotiation kept human; a little wary of usability.
IP2	Industry Peer (young professional)	Tea Taster / Grader	5–10 yrs	Tastes and grades using personal judgement and taste memory. Genuinely positive about AI in principle, though not confident it could match human judgement on tea; would welcome help with record-keeping, buyer matching, price guidance, tasting notes and general decision support, provided aroma and taste stay a human call. Accuracy and reliability are his main concerns.

continued on next page

Table V.1 – continued from previous page

Code	Group	Role	Experience	Relevant background
IP3	Industry Peer (experienced)	Owner / Proprietor	20+ yrs	Covers the whole process himself: tasting, buying, negotiating, buyer relationships and record-keeping. Most concerned with knowledge transfer: staff with five or six years' experience still "cannot tell me why they gave a grade," and would rather have a tool that helps train people than one that grades for them. Sees the auction catalogue system as overdue an update, but insists the final call must stay with the person who answers to the buyer.
IP4	Industry Peer (experienced)	Tea Trader	16–20 yrs	Broadest role of anyone interviewed: tasting, buying, negotiating, buyer relationships, plus auction bidding and blend approval for packet orders. Talks about the work in sensory terms: "colour can be corrected in a blend, smell cannot," and "half of what I judge never reaches the eye." Wants a record of who tasted a sample and when; more concerned about mismatches between the auction sample and delivered chest than grading disagreements. Open to AI for records and price history, but clear that grading itself should stay human.

V.2 Design Validation Focus Group

Table V.2: Focus group session details

Group	Participants	Format	Purpose	Data form
Case-company (Shree Shyaam Tea Co.)	P1, E1, E2	Joint call	Validate the proposed platform design (Chapter 6)	One combined researcher note; no audio or video recording

The focus group involved the same three case-company participants who completed the individual interviews. It represents a second data-collection stage, not three additional participants. The unique participant count for the study remains seven.

Appendix VI: Full Coding Matrix

The table below lists every coded meaning unit from the seven interviews, grouped by final theme, with the alignment call applied during validation.

Alignment key: **S** = Supports the theme directly. **P** = Partial, coded to the theme's category but lighter-weight or tangential. **C** = Contradicts, the meaning unit runs counter to the theme as worded and is retained deliberately rather than filtered out.

Ppt = Participant code (see Appendix V for role and experience)

Verification note: this matrix has been checked line by line against the raw questionnaire responses (University of Southampton Research Questionnaire). A first pass had omitted several coded units that Chapter 4 and the cross-group comparison rely on directly, most notably the majority position on Theme 4 and the full spread of participants on the Theme 1 fixed-versus-individual split. Those units have been added below so that every claim made in the Findings chapter now has a traceable row in this appendix. One factual correction has also been made to the Theme 6 entry for P1 (documentation practice), which had not accounted for P1's own answer on digital-system use.

VI.1 Theme 1 — Grading is experiential and context-dependent, not standardised

Table VI.1: Coded evidence, Theme 1

Ppt	Evidence	Align
P1	Poor lighting and environmental conditions cited as main cause of inconsistent assessment; morning natural light used deliberately.	S
P1	No single fault disqualifies a lot alone, decision depends on the overall picture. <i>(Holistic judgement; see row below on P1's fixed-standard framing.)</i>	S

continued on next page

Table VI.1 – continued

Ppt	Evidence	Align
P1	Describes grade labels (e.g. BOP, PF) as "a fixed industry standard", while noting that aroma expectations can shift the meaning of the same label between gardens.	C
E1	Describes grade labels (e.g. BOP, PF) as "a fixed industry standard." <i>(One of the sharpest textual contradictions in the dataset.)</i>	C
E1	Off/musty aroma and weak/thin liquor named as disqualifying faults.	S
E1	Weighs colour and aroma equally rather than favouring one.	S
E2	Describes grade labels (e.g. BOP, PF) as "a fixed industry standard", identical wording to P1 and E1. <i>(Completes the case-company set; all three respond identically to this question.)</i>	C
E2	Quality score translates "very directly" into quoted price. <i>(Related to consistency, not the fixed-vs-individual question itself; P1 gives the identical answer.)</i>	P
IP1	"I mainly rely on my own judgement, regardless of the label."	S
IP1	Reports longest per-sample assessment time in the sample (5+ minutes).	S
IP2	"I mainly rely on my own judgement, regardless of the label." <i>(Identical wording to IP1 and IP4.)</i>	S
IP2	When colour and aroma conflict, "aroma usually wins."	S
IP2	Names humidity, rather than lighting, as the condition that makes visual assessment hardest, the only participant not to cite poor lighting here.	S
IP3	Grade meaning "varies significantly by garden or region" rather than being fixed.	S
IP4	"I mainly rely on my own judgement, regardless of the label." <i>(Identical wording to IP1 and IP2.)</i>	S
IP4	"Colour can be corrected in a blend, smell cannot."	S

VI.2 Theme 2 — Trust in AI is conditional and evidence-based

Table VI.2: Coded evidence, Theme 2

Ppt	Evidence	Align
P1	Never encountered AI in food/agriculture quality assessment before this study.	S

continued on next page

Table VI.2 – continued

Ppt	Evidence	Align
P1	Stacks multiple trust conditions before accepting a grade: an explanation for the output, a comparison against a past sample, and the ability to override it.	S
E1	Trust requires the ability to review and override the AI's output.	S
E2	"Very positive" about AI; believes it could reduce disputes with buyers. (<i>Sentiment, not a specific evidentiary bar.</i>)	P
E2	Would need "a clear reason or explanation for the grade given" before trusting it.	S
E2	Had already heard of AI in food/agriculture quality assessment before this study. (<i>Background, not a stated trust condition.</i>)	P
IP1	Trust requires "accuracy proven over time" rather than an explanation alone.	S
IP2	"Positive" about AI overall, yet "not confident it can match human judgement for tea."	S
IP3	Would want to "test it beside with me for a month or so" before trusting a grade.	S
IP3	Also ties trust to a full stacked set of conditions beyond the trial period: an explanation for the grade, a comparison to a recognised past sample, override rights, and accuracy proven over time.	S
IP3	Selects all five listed concern types at once: accuracy, cost, privacy, reliability, trust. (<i>IP4 also selects all five; not unique to IP3.</i>)	S
IP4	Also selects all five listed concern types at once: accuracy, cost, privacy, reliability, trust.	S
IP4	Stacks the same four conditions as IP3 before trusting a grade: explanation, comparison to a past sample, accuracy proven over time, override rights.	S
IP4	"Taste is in the mouth and aroma is in the nose, and no camera has either." (<i>A categorical limit on what evidence could ever satisfy, not a threshold the technology simply has not reached yet, distinct from the conditional trust everyone else in this table describes.</i>)	C

VI.3 Theme 3 — AI's realistic value lies in support functions, not sensory judgement

Table VI.3: Coded evidence, Theme 3

Ppt	Evidence	Align
P1	Wants a wide AI feature set spanning functional benefits (image analysis, buyer matching, price guidance, auto-generated notes) and platform features (lighting correction), the broadest wish-list alongside IP3's.	S
E1	Names record keeping and matching lots to buyer preferences as areas AI could help with, while keeping judging aroma or taste as strictly human.	S
IP1	Names matching tea lots to buyer preferences as an AI-suitable function, and frames AI's main value as helping "preserve tea-tasting knowledge for the future" rather than grade in place of a person.	S
IP2	Broadest set of AI benefit areas of any young professional: records, lot matching, price guidance, notes, decision support.	S
IP3	Selects the widest feature set of any participant overall (six items, including historical comparison and lighting correction alongside grading support), wider than P1's five.	S
IP3	Wants a system that "sits with" trainees and shows comparisons, rather than one that grades for them.	S
IP4	"Half of what I judge never reaches the eye." Doubts a camera-based system can capture what matters. <i>(Framed as a categorical limit, not a boundary more evidence could shift.)</i>	C
IP4	Only accepts AI for historical comparison, records, and tasting notes; explicitly excludes it from grading itself.	S

VI.4 Theme 4 — Human judgement and accountability must remain central

Table VI.4: Coded evidence, Theme 4

Ppt	Evidence	Align
P1	Would review and override an AI grade silently rather than escalate or discuss it.	S
E1	If AI disagreed with a grading, wants it to "learn from my correction" rather than a one-off override.	S
E2	If AI disagreed with a grading, would override it silently.	S
IP1	If AI disagreed with a grading, wants it to "learn from my correction."	S
IP2	If AI disagreed with a grading, wants it to "learn from my correction."	S

continued on next page

Table VI.4 – continued

Ppt	Evidence	Align
IP3	If AI disagreed with a grading, wants it to "learn from my correction." <i>(The majority position: five of seven participants land here, E1, IP1, IP2, IP3, and, per Theme 2, the same correction-based framing sits alongside IP3's separate accountability statement below.)</i>	S
IP3	"The final call must stay with the person who has to answer to the buyer."	S
IP3	Names judging unusual or unfamiliar lots as something that should always stay human.	S
IP4	If AI disagreed, prefers to "flag it for review" rather than silently override.	S

VI.5 Theme 5 — Data control and cost sensitivity shape adoption willingness

Table VI.5: Coded evidence, Theme 5

Ppt	Evidence	Align
P1	Would only accept a small one-time cost (e.g. equipment), not a subscription.	S
E1	Comfort with cloud storage "depends what is stored"; prefers data staying within India / the business.	S
E1	Willing to invest significantly, but only once a tool proves its value, not up front.	S
E2	A tool would need to be free, "no cost."	S
IP1	Prefers data stored locally, citing a general privacy preference.	S
IP1	Open to a small monthly subscription rather than a one-off payment. <i>(Privacy-cautious but cost-open, breaks the assumption these two move together.)</i>	P
IP2	Concerned about competitors accessing data, which drives their preference for local-only storage.	S
IP2	Willing to invest significantly, but only once a tool proves its value.	S
IP3	Comfort with cloud storage "depends what is stored"; reasoning cites both competitor access and a general privacy preference.	S
IP3	Would only accept a small one-time cost (e.g. equipment).	S
IP4	Prefers data stored locally only; reasoning cites both a general privacy preference and concern about competitors accessing data.	S

continued on next page

Table VI.5 – continued

Ppt	Evidence	Align
IP4	Would only accept a small one-time cost (e.g. equipment), in the same bracket as P1 and IP3.	S

VI.6 Theme 6 — The deeper business problems lie beyond grading itself

Table VI.6: Coded evidence, Theme 6

Ppt	Evidence	Align
P1	Wants AI to capture "general experience / intuition, hard to put into words" so the business is not dependent on one person.	S
P1	Documentation kept mainly as personal notes and paper records, supplemented by basic Excel use, not a dedicated digital grading or record-keeping system. <i>(Corrected: P1 does report using Excel elsewhere in the questionnaire, so "no digital system at all" would overstate the point.)</i>	S
E1	Records are documented in Excel <i>(per the record-keeping question)</i> , though E1 separately answers "no" when asked directly about using any digital system, an inconsistency in E1's own responses worth noting rather than silently resolving.	S
E2	Volunteers, unprompted, that "blending is very important part" of the process.	S
IP1	Sees AI's main value as helping "preserve tea-tasting knowledge for the future."	S
IP1	Already uses Excel plus a separate digital system for tasting/grading records.	S
IP2	Buyer preferences currently managed with "no formal system."	S
IP3	"I have people with five, six years who still cannot tell me why they gave a grade."	S
IP4	Wants the platform to store "who tasted it and on what date, so I can go back and check."	S
IP4	"Sample quality from the auction side... that is a bigger loss to me than any grading disagreement."	S
IP4	Does not select "intuition hard to put into words" as knowledge worth capturing, unlike P1 and IP3. <i>(Excludes tacit knowledge, unlike the two owners.)</i>	C

Appendix VII: Focus Group Design Validation Evidence

This appendix records the evidence gathered in the design validation focus group conducted with the three case-company participants (P1, E1, E2) of Shree Shyaam Tea Co. The session took place as a joint call after the proposed platform design, set out in Chapter 6, had been explained to participants. No audio or video recording was made; responses are documented as a single combined researcher note and cannot be attributed to individual participants. This appendix is separate from Appendix VI, which covers Stage 1 interview data only. The focus group did not generate themes. Its purpose was to test whether the finished design met the needs participants had described in Stage 1.

Table VII.1: Focus group responses and design validation implications

Q	Question	Response	Validation implication
Q1	Overall, how useful does this proposed system look for your business?	Somewhat useful	Conditional endorsement. The platform is accepted as useful in principle, though not enthusiastically. This is consistent with the cautious openness recorded across Themes 2 and 3 in Stage 1 and supports the tiered adoption model proposed in Section 6.3.2.
Q2	Which part would provide the biggest benefit?	Faster sample preparation and image taking	Participants identify workflow efficiency rather than grading quality as the primary benefit. This confirms the design priority placed on the two-tray preparation system and the structured image-capture step in Section 6.3.

continued on next page

Table VII.1 – continued from previous page

Q	Question	Response	Validation implication
Q3	Does the two-tray system make the tasting workflow more practical?	Yes	Direct validation of the two-tray design. Supports retaining this feature as specified in Section 6.3.
Q4	Should the taster remain responsible for the final sensory judgement?	Yes, but AI can assist with colour analysis and tea particle grading analysis if taught with data	Confirms the human-in-the-loop boundary established in Stage 1 and maintained throughout Chapter 5. Participants accept AI assistance for colour and particle assessment while keeping the overall sensory verdict with the taster. This directly supports the design boundary discussed in Section 6.1.
Q5	Which AI function would you be most comfortable using?	Note generation and Image analysis	Both functions sit on the support side of the AI-human boundary. Selection of these two over grading or price-guidance functions reinforces the finding from Theme 3 that participants draw AI's realistic value from documentation and visual analysis rather than sensory judgement.
Q6	Would you be comfortable with the proposed digital record containing images, tasting notes, buyer history and sales history?	Yes, if stored locally or securely	The conditional acceptance replicates the data-locality preference recorded under Theme 5. Local or secure storage is not a negotiable preference but a stated requirement, and the platform design in Section 6.3.2 must reflect this.

continued on next page

Table VII.1 – continued from previous page

Q	Question	Response	Validation implication
Q7	If the system worked as shown, would you consider using it?	Yes	Unqualified adoption intent. The only unambiguously positive response in the set. Confirms that the design as explained is acceptable to the case-company participants as a whole.
Q8	Is there anything in the proposed system you would change or remove?	"Yes add some features integral to the process but would know after trial and error of the feature we do manually but feel difficult with the machine"	Participants anticipate friction between current manual practice and the system as designed, but cannot specify which functions until use begins. This supports the recommendation in Section 6.3.3 for a phased trial period rather than full deployment from the outset.

The responses above are referenced in Sections 6.1, 6.3, and 6.4 when assessing whether the proposed design aligns with what participants said they need.

Appendix VIII: AI Declaration

Several AI tools supported this research, each in a limited and specific way. ChatGPT was used for early brainstorming and image generation. NotebookLM helped process large volumes of research papers. Perplexity AI and Google Scholar were used for literature searching. Claude assisted with structuring and organising written material.

None of these tools produced the arguments or conclusions in this dissertation. That work is the author's own. AI outputs were reviewed and rewritten throughout; nothing was accepted without independent judgement.